



A comprehensive survey on robust federated learning: From threat models to defense mechanisms

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ABSTRACT

Federated Learning (FL) enables collaborative model training without centralizing raw data and has become an important paradigm for privacy-sensitive intelligent systems. Nevertheless, its distributed architecture, heterogeneous data, and unreliable participants introduce substantial robustness and security challenges, ranging from communication noise and system failures to poisoning, privacy inference, and Byzantine attacks. In this paper, we present a structured survey of robust federated learning. We first organize the literature through a threat-oriented taxonomy that covers poisoning attacks, privacy leakage risks, Byzantine behavior, and scenario-specific robustness challenges. We explicitly analyze stronger adversaries that collude, coordinate poisoning campaigns, replace models behind secure aggregation, and adapt their optimization to defense-specific weaknesses. We then review representative defense mechanisms, including robust aggregation, backdoor defense, privacy-preserving training, secure aggregation, and personalized robustness enhancement. Particular attention is given to three recurring tensions in the literature: robustness under Non-IID data, robustness-efficiency trade-offs, and robustness-privacy trade-offs. We further discuss robust federated learning in wireless and IoT environments, where communication uncertainty and resource constraints become first-order concerns. We examine how existing robustness mechanisms translate, or fail to translate, to foundation-model-based FL, federated fine-tuning of large language models, federated reinforcement learning, agentic federated systems, and federated retrieval-augmented generation. Finally, we summarize open challenges and outline future research directions for building more reliable, efficient, and deployable federated learning systems.

1. Introduction

Nowadays, 5G networks, smart devices, and the Internet of Things (IoT) have become essential components of daily life. The use of these devices and application services generates massive amounts of data. Contemporary technological development heavily relies on data; however, centralized data collection requires stable network connections and high-speed broadband, which often conflicts with the resource constraint devices. Furthermore, maintaining data integrity and privacy is of utmost importance. The transfer of sensitive data related to individuals, organizations, and governments carries significant security risks.

Unlike centralized data collection algorithms, federated learning is a distributed machine learning paradigm designed to enable collaborative model training by multiple parties while avoiding the centralized sharing of raw data. Its core idea is that each participating party, such as user devices, medical institutions, and enterprises, trains a model locally and only uploads model parameters or updates to a central server for aggregation, thereby balancing data privacy protection with model performance.

The federated learning framework inherently supports security and privacy because data generated on terminal devices do not need to leave those devices. Useful local data are used to train models in a distributed

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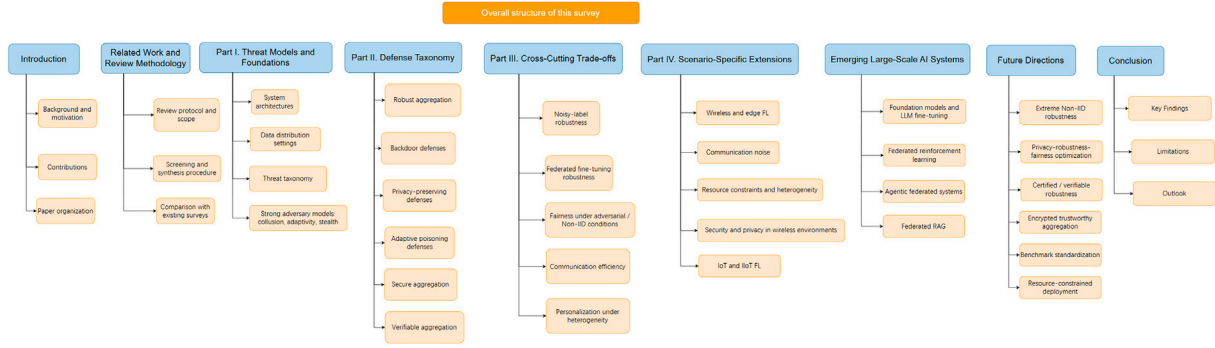


Fig. 1. Overall structure of this survey.

manner, and only model updates are exchanged between end devices and the server. However, this design still exposes important security threats. These threats may degrade model utility, violate participant privacy, and emerge at different stages of the federated learning life cycle. Recent surveys show that attacks against federated learning have become increasingly diverse, covering poisoning, privacy inference, and system-level vulnerabilities, which in turn have motivated a broad range of defense mechanisms [33,39,48].

Typical application scenarios for federated learning include cross-institutional healthcare analytics, mobile and edge intelligence, and security-sensitive IoT services [16,25,40]. Furthermore, federated learning has been explored in a wide range of environments, such as smart cities, wireless communications, edge networking, industrial monitoring, and real-time distributed denial-of-service (DDoS) detection [23, 25,40]. Recent work has also extended this paradigm to emerging settings such as wireless federated learning and quantum federated learning [47].

Research on federated learning has received significant attention in recent years, but a central challenge remains: how to jointly balance privacy, robustness, model performance, and system efficiency. To make this literature easier to navigate, this paper restructures robust federated learning around four main parts: threat models, defense taxonomies, cross-cutting trade-offs, and scenario-specific extensions. Based on this structure, we also identify important open problems and future directions.

Our contributions are summarized as follows:

1. We provide a structured overview of federated learning architectures, data-distribution settings, and threat surfaces, and summarize how these design choices shape robustness risks.
2. We review representative attacks and defense mechanisms through a threat-oriented taxonomy, explicitly distinguishing independent attackers from colluding, coordinated, stealthy, and defense-aware adaptive adversaries and identifying the assumptions these stronger attackers invalidate.
3. We analyze cross-cutting trade-offs among secure aggregation, privacy protection, robustness enhancement, communication-efficient training, fairness, and personalization.
4. We extend the discussion to wireless, edge, and IoT scenarios, where communication noise, resource constraints, and system heterogeneity reshape both the threat surface and the design of defenses.
5. We analyze emerging large-scale AI settings, including foundation models, large language model (LLM) fine-tuning, federated reinforcement learning, agentic systems, and federated retrieval-augmented generation (RAG), and identify where update-level defenses fail to address their expanded attack surfaces.
6. We provide a forward-looking research roadmap that transforms recurring limitations into concrete open problems, including

extreme non-independent and non-identically distributed (Non-IID) robustness, joint privacy–robustness–fairness optimization, certified guarantees, encrypted trustworthy aggregation, benchmark standardization, and resource-constrained edge deployment.

The remainder of this paper is organized as follows. [Section 2](#) describes the review methodology and compares related surveys. [Section 3](#) presents Part I, which introduces federated learning foundations and a threat-oriented view of robustness. [Section 4](#) presents Part II, which organizes representative defense families, including robust aggregation, privacy-preserving defenses, secure aggregation, and verifiable aggregation. [Section 5](#) presents Part III, which analyzes trade-offs among robustness, privacy, communication efficiency, fairness, and personalization. [Section 6](#) presents Part IV, which extends the discussion to wireless and IoT environments. [Section 7](#) analyzes emerging large-scale AI systems, [Section 8](#) presents future research directions, and [Section 9](#) concludes the paper.

The structure of survey is shown in [Fig. 1](#).

2. Related work and review methodology

To position this survey within the existing literature, we first describe the review protocol and then compare recent surveys on federated learning security, privacy, robustness, fairness, and communication efficiency. Compared with prior surveys, our emphasis is on robustness-oriented defense mechanisms and on the interactions among security, privacy, heterogeneity, and efficiency in federated learning.

2.1. Review protocol and scope

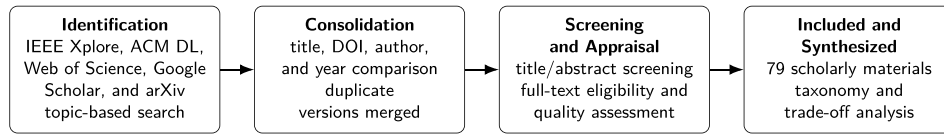
The literature collection and synthesis followed a structured review protocol informed by Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) reporting concepts [44]. The information sources used for literature discovery were IEEE Xplore, ACM Digital Library, Web of Science, Google Scholar, and arXiv. The principal publication window was January 2020 through December 2025; selected earlier foundational studies were included when they established concepts required by the survey, including federated averaging (FedAvg), sign-based gradient quantization, Top- k sparsification, gradient leakage, secure aggregation, Byzantine-robust aggregation, and machine-unlearning-based privacy lifecycle management.

Searches across the five information sources used combinations of federated-learning terms with threat, defense, system-property, and deployment terms. The normalized core query was adapted to the Boolean and wildcard syntax supported by each information source:

(“federated learning”) AND (robust* OR secur* OR attack* OR poison* OR backdoor OR Byzantine OR collud* OR coordinat* OR adaptive OR “model replacement” OR “privacy inference” OR

Table 1
Structured review protocol.

Protocol Item	Specification
Review objective	Identify and synthesize threat models, robustness-oriented defenses, privacy mechanisms, system trade-offs, and deployment challenges in federated learning.
Information sources	IEEE Xplore, ACM Digital Library, Web of Science, Google Scholar, and arXiv; publisher records and original full texts were used for bibliographic verification, with DOI/arXiv metadata independently cross-checked.
Search scope	Source-appropriate combinations of “federated learning” with robustness, security, attack, poisoning, backdoor, Byzantine, collusion, coordination, adaptivity, model replacement, privacy, secure aggregation, verifiability, certification, benchmark, heterogeneity, fairness, personalization, unlearning, explainability, threat detection, cyber-physical-social systems, wireless, IoT, communication efficiency, gradient compression, quantization, sparsification, signSGD, Top- <i>k</i> , foundation models, large language models, reinforcement learning, agents, and retrieval-augmented generation.
Temporal and language scope	English-language studies published mainly from January 2020 to December 2025, plus selected pre-2020 foundational sources and 2026 material needed to explain core federated-learning, privacy, and robustness concepts.
Inclusion criteria	Full-text studies directly relevant to the review objective; peer-reviewed studies were prioritized, while highly relevant arXiv studies were included when final publication metadata was unavailable.
Exclusion criteria	Duplicate or superseded versions; studies without a direct federated-learning connection; papers only tangentially related to the taxonomy; and records without sufficient technical detail for analysis.
Screening and deduplication	Records were consolidated using title, DOI, author, and publication-year information. Title and abstract screening was followed by full-text eligibility assessment against the inclusion and exclusion criteria.
Quality assessment	Eligible studies were assessed qualitatively for relevance, clarity of the threat or system model, technical completeness, empirical or theoretical support, reproducibility information, and explicitly reported limitations.
Data extraction and synthesis	For each included study, the analysis considered its threat or problem setting, core mechanism, assumptions, evaluation setting, reported benefits, trade-offs, and limitations. The extracted evidence was synthesized into the taxonomy used in Parts I–IV.
Final corpus and audit trail	The qualitative synthesis includes 79 scholarly publications or official scholarly materials. Two additional contextual sources describe the UCI and COMPAS datasets, and one methodological publication informs PRISMA-style reporting. Bibliographic metadata and manuscript citations were reconciled through a citation-keyed corpus manifest.

**Fig. 2.** PRISMA-style literature selection and synthesis workflow used in this survey.

“secure aggregation” OR verif* OR certif* OR benchmark OR heterogeneous* OR fairness OR personali* OR unlearn* OR explainable OR “threat detection” OR “cyber-physical-social” OR wireless OR IoT OR “communication efficiency” OR compress* OR quantiz* OR sparsif* OR signSGD OR “top-*k*” OR “foundation model” OR “large language model” OR “reinforcement learning” OR agent* OR “retrieval-augmented generation”).

The asterisk denotes a truncation wildcard for common word variants, such as “verification” and “verifiable.” The query combines these term families because robustness in federated learning spans both adversarial and non-adversarial failure modes.

2.2. Screening results and synthesis procedure

Records retrieved from the five information sources were consolidated by comparing titles, digital object identifier (DOI) information, author lists, publication years, and available full texts. Title and abstract screening identified studies aligned with the survey scope, after which full texts were assessed using the eligibility and quality criteria in Table 1. Duplicate and superseded versions were consolidated, with the most complete publication record used for citation.

The resulting review corpus contains 79 scholarly publications or official scholarly materials included in the qualitative synthesis. Each included publication was mapped to the survey taxonomy using its threat or problem setting, defense mechanism, assumptions, evaluation evidence, benefits, trade-offs, and limitations. Two additional contextual sources are cited only to describe the UCI Adult and COMPAS datasets, and the PRISMA publication is cited as a methodological reference. Fig. 2 summarizes this PRISMA-style selection and synthesis workflow.

2.3. Comparison with existing surveys

Recent surveys commonly focus on a particular threat family, system property, or deployment setting. Table 2 compares their coverage with the broader robustness-oriented scope adopted in this paper.

Representative primary studies also show why robust FL cannot be evaluated through aggregation accuracy alone. Adaptive model-poisoning defenses such as FedDet emphasize layer-wise behavior and defense-aware attackers, while privacy-preserving and verifiable mechanisms such as MASKCRYPT, TAPFed, and VERSA show that encryption, secure aggregation, and protocol auditing reshape what the server can inspect or prove [26,27,71,72]. Recent encrypted-gradient auditing work further illustrates this tension by combining additive homomorphic encryption with Gaussian mixture modeling and Mahalanobis-distance analysis to detect model poisoning without exposing plaintext gradients [75]. In deployment-oriented settings, personalized and edge-aware defenses such as Sylva and DSPFL address heterogeneity and resource constraints, while explainable cyber-physical-social systems (CPSS) threat-detection work combines FL with differential privacy, Paillier-based partially homomorphic encryption, and SHapley Additive exPlanations (SHAP) explanations to connect robustness with reliability and operator trust [46,70,76]. Finally, post-training privacy mechanisms such as TensorUnlearn broaden the robustness discussion beyond training-time aggregation by highlighting efficient approximate unlearning and local-regime membership-inference-risk bounds after model deployment [49]. Taken together, these primary studies motivate our emphasis on compatibility among robust aggregation, cryptographic protection, explainability, deployment context, and post-training privacy requirements.

3. Part I: threat models and system foundations

This part establishes the system assumptions that determine how threats arise in federated learning. We first summarize the architectural

Table 2

Comparison of recent surveys on federated learning security, privacy, and robustness (2024–2025).

Ref.	Core Perspective	Attacks		Defenses			Properties	Specific Scenario
		Poisoning	Inference	Robust Aggr.	Privacy (DP/HE)	Verifiability	Fairness/ Gen.	
Li et al. [39]	Full Life Cycle Threats	●	●	●	●	●	●	General
Wan et al. [62]	Wireless & Backdoor	●	○	●	●	○	○	Wireless FL
Atia et al. [2]	IoT Security Frameworks	●	●	●	●	○	○	IoT / Edge
Kumar et al. [33]	Impact Analysis	●	●	●	○	○	○	General
Rao et al. [48]	Privacy Inference	○	●	○	●	○	○	Centralized & FL
Fouda et al. [22]	Privacy Preserving Tech	○	●	○	●	○	○	Emerging Networks
Huang et al. [28]	Performance Properties	●	○	●	○	○	●	Non-IID Data
Xing et al. [69]	ZKP & Verifiability	○	○	○	●	●	○	Decentralized Net
Ours	Robust Defence Frameworks	●	●	●	●	●	●	

Legend: ●: High coverage/Focus; ○: Medium coverage/Mentioned; ○: Low coverage/Not involved. *Abbreviations:* Aggr.: Aggregation; DP: Differential Privacy; HE: Homomorphic Encryption; Gen.: Generalization; ZKP: Zero-Knowledge Proof.

and data-distribution settings of federated learning, and then organize the main robustness threats according to how they affect confidentiality, integrity, and system reliability. The aim is to show that robustness problems are inseparable from deployment assumptions such as centralized versus decentralized coordination, cross-device versus cross-silo participation, and the severity of statistical heterogeneity. The typical architecture of federated learning consists of three components: a central server, clients, and a communication channel. The central server coordinates each training round by aggregating local model updates and distributing the latest global model. Clients independently train on their local data and upload model parameters or gradients through the communication channel. Because this channel is responsible for transmitting learning signals rather than raw data, its design must jointly consider privacy, integrity, and communication efficiency [40,61]. This distributed architecture helps alleviate the tension between data utility and centralized data collection.

A commonly used empirical risk minimization objective in federated learning, popularized by FedAvg, is written as follows [43]:

$$f(w) = \sum_{k=1}^K \frac{n_k}{n} F_k(w) \quad \text{where} \quad F_k(w) = \frac{1}{n_k} \sum_{i \in P_k} f_i(w). \quad (1)$$

In Eq. (1), $f_i(w)$ represents the prediction loss function for input x_i to desired output y_i , with weight vector w , K is the number of participants in the current learning round, and $F_k(w)$ is the local objective function of the k -th participant. For the total number of samples n , n_k is the number of samples locally available to the k -th participant. Similarly, P_k with $n_k = |P_k|$ is the partition assigned to the k -th participant from the entire dataset P .

In a typical federated learning setting, devices first download the current model parameters from the server, initialize the local model, and then train on their local datasets. The parameters are optimized by minimizing the local objective function, and the resulting local updates are aggregated on the server using FedAvg [28,43]. In this manner, the global model is updated round by round.

Since raw data remains on local devices and only model updates are transmitted, federated learning reduces direct data exposure. However, it does not guarantee privacy by itself. In practice, privacy protection is often strengthened through techniques such as differential privacy, secure aggregation, secure multi-party computation, and homomorphic encryption [9,22,48].

Based on system architecture and data distribution characteristics, federated learning can be divided into the following three categories:

3.1. Foundational system architecture settings

- **Centralized Federated Learning:** Relies on a single central server to coordinate training and aggregate client updates. Its advantages include simple implementation and high aggregation efficiency, but the central server introduces a trust bottleneck and

a single point of failure. Centralized federated learning is shown in Fig. 3.

- **Decentralized Federated Learning:** Model updates are exchanged directly through a peer-to-peer (P2P) network, reducing the risk of a single point of failure but facing higher communication overhead [69]. For example, in IoT scenarios, edge devices can communicate directly with other devices without relying on a central server. The decentralized federated learning is shown in Fig. 4.
- **Hybrid Federated Learning:** Combines centralized and decentralized architectures, for instance, aggregating model parameters hierarchically through edge servers to balance efficiency and robustness [31].

3.2. Foundational data distribution settings

- **Horizontal Federated Learning (HFL):** Participants share the same feature space but hold different samples, such as customer data from different banks. HFL expands the available sample population without requiring participants to share raw data. Typical applications include cross-institutional modeling, such as multiple hospitals collaboratively training a disease diagnosis model [59].
- **Vertical Federated Learning (VFL):** Participants share the same samples but different features, such as a hospital and an insurance company collaborating to analyze medical records and insurance records of the same patients. This mode is suitable for scenarios with severe data silo problems.
- **Federated Transfer Learning (FTL):** Leverages knowledge from a source domain to improve learning in a target domain when participants have limited overlap in both samples and features. For example, hospitals with different patient populations and clinical feature spaces can transfer knowledge to improve diagnostic performance at a data-scarce target hospital.

The comparison of horizontal federated learning, vertical federated learning, and federated transfer learning is shown in Fig. 5.

3.3. Comparison of foundational settings

The comparison of federated learning classifications is shown in Table 3. The table organizes federated learning along two complementary axes: system architecture and data distribution. The architecture dimension explains where coordination and aggregation occur, contrasting the efficiency and simplicity of centralized FL with the fault tolerance of decentralized FL and the layered trade-offs introduced by hybrid FL. The data-distribution dimension focuses on how samples and features are shared across participants: HFL expands the sample population under a common feature space, VFL combines complementary features for overlapping entities, and FTL supports collaboration when both sample and feature overlap are limited. These distinctions are important for robust FL because they determine

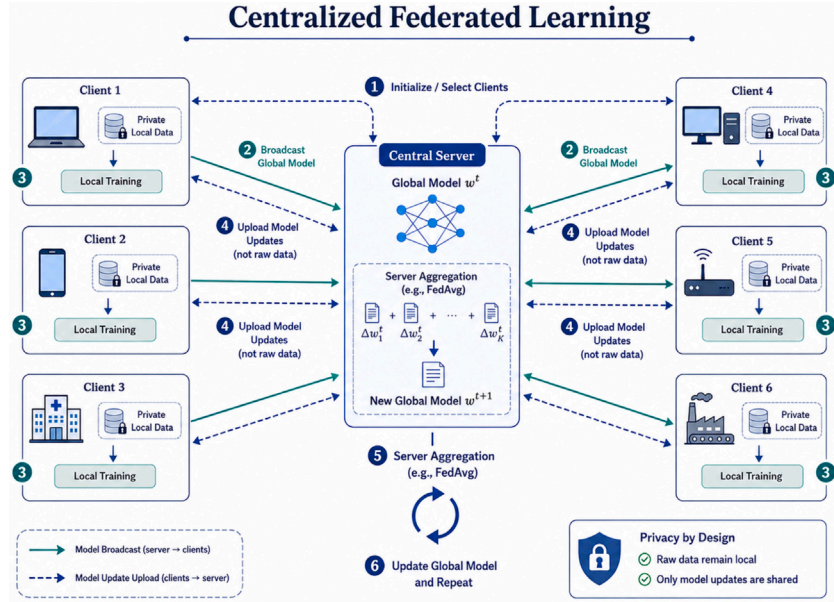


Fig. 3. Centralized federated learning.

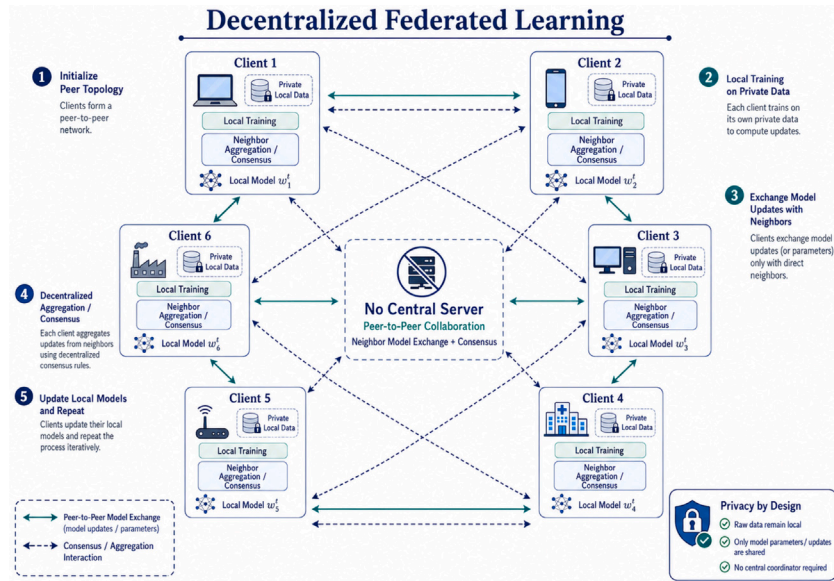


Fig. 4. Decentralized federated learning.

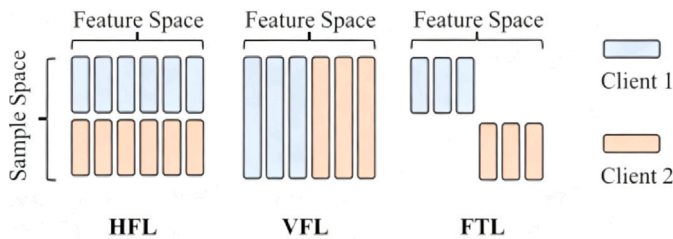


Fig. 5. Comparison of horizontal federated learning, vertical federated learning, and federated transfer learning.

the main trust assumptions, communication bottlenecks, privacy constraints, and heterogeneity patterns that later defense mechanisms must address.

3.4. Threat taxonomy in robust federated learning

Despite its significant advantages in privacy protection, the practical deployment of federated learning still faces multiple challenges. From a survey perspective, these threats can be grouped into confidentiality threats, integrity threats, and system-level robustness threats. This grouping also prepares the transition to Part II, where the corresponding defense families are discussed.

- **Privacy Leakage Risks:** Model updates may still reveal sensitive information through attacks such as membership inference and gradient inversion [48,81].
- **Byzantine and Poisoning Attacks:** Malicious clients may submit forged or manipulated updates to disrupt convergence or implant targeted behaviors in the global model. Stronger attackers may coordinate several clients, optimize against the deployed aggregation

Table 3

Comparison of federated learning classifications with representative examples.

Classification Dimension	Type	Representative Example	Advantage	Disadvantage
System Architecture	Centralized	Cross-device keyboard prediction, where smart-phones train locally and a cloud server aggregates their updates.	Efficient coordination and relatively simple deployment.	The central server creates a trust bottleneck and a single point of failure.
	Decentralized	Peer-to-peer IoT anomaly detection, where neighboring devices exchange and aggregate model updates without a central server.	Eliminates dependence on a single coordinator and improves fault tolerance.	Requires costly peer-to-peer communication and consensus mechanisms.
	Hybrid	Smart-city or vehicular learning, where devices send updates to edge servers for regional aggregation before cloud aggregation.	Balances scalability, communication efficiency, and global coordination.	Introduces architectural complexity and additional edge-server trust assumptions.
Data Distribution	Horizontal	Hospitals using the same clinical feature schema but different patient records to collaboratively train a diagnosis model.	Expands the available sample population without sharing raw data.	Non-IID sample distributions may cause client drift and unstable convergence.
	Vertical	A bank and an e-commerce company jointly training a credit-risk model for shared customers using different attributes.	Combines complementary features held by different organizations.	Requires entity alignment and costly privacy-preserving computation.
	Federated Transfer Learning	Hospitals with limited overlap in both patients and feature spaces transferring knowledge to a data-scarce target hospital.	Supports collaboration when both sample and feature overlap are limited.	May suffer from negative transfer when the source and target domains differ substantially.

Table 4

Threat taxonomy in robust federated learning.

Threat Family	Representative Threats	Primary Asset at Risk	Typical Adversarial Capability	Representative Sources
Confidentiality threats	Membership inference, property inference, gradient inversion, and model extraction	Training-data privacy and client confidentiality	Access to model outputs, gradients, or aggregated updates	[39,48,81]
Integrity threats	Byzantine updates, data poisoning, model poisoning, and backdoor insertion	Global model correctness and prediction behavior	Malicious local training, forged gradients, and collusion across clients	[4,21,39,52,66]
Heterogeneity-amplified threats	Distribution shift, Non-IID-induced false alarms, and subgroup unfairness	Robust generalization and stable aggregation	Exploiting statistical skew or client drift to hide malicious behavior	[28,33,53]
System-level threats	Communication bottlenecks, unreliable channels, aggregator failure, and replay or collusion	Availability, scalability, and reliable protocol execution	Exploiting bandwidth limits, lossy links, or weak aggregation assumptions	[6,61,71]

rule, or preserve clean accuracy while inserting a stealthy backdoor [4,21,39,52,66].

- **Adversarial Robustness Under Heterogeneity:** Data heterogeneity and distribution shifts can amplify the difficulty of maintaining robustness across clients and scenarios [28,33].
- **Communication Efficiency:** Repeated transmission of high-dimensional model updates imposes substantial bandwidth and latency costs, especially in edge and wireless settings [6,61].

Table 4 summarizes this threat-oriented view and highlights the main assets at risk, typical adversarial capabilities, and the system assumptions under which these threats become most severe.

3.4.1. Stronger adversary models: collusion, adaptivity, and stealth

A malicious-client ratio alone does not adequately characterize the adversary. A robust federated learning evaluation should additionally specify at least four capability axes: (i) whether malicious clients act independently or share updates, data, triggers, and timing; (ii) what the attacker knows about the aggregation rule, defense thresholds, benign-update distribution, and global model; (iii) whether individual updates are visible to the server or hidden by secure aggregation; and (iv) whether the attack is a one-round event or adapts across rounds. These axes determine whether common assumptions such as independent failures, geometric separability, benign-majority behavior, server-side inspectability, and stationary attack distributions are credible.

Collusion changes the statistical structure observed by a defense. Instead of producing isolated outliers, a coalition can submit mutually consistent updates, distribute a trigger across several clients, or coordinate its timing and scale so that no single contribution appears decisive. The distributed backdoor attack (DBA), for example, decomposes a global trigger across multiple compromised clients, while optimized

model-poisoning attacks coordinate malicious updates against the target aggregation rule [21,52,66]. The relevant quantity is therefore not only the number of malicious clients, but also their ability to behave as a coherent subgroup.

Secure aggregation introduces a particularly important robustness-privacy tension. It protects individual updates from inspection, but secrecy does not establish that the hidden updates are benign. Model-replacement attacks exploit this gap by scaling a malicious local update so that the aggregate is driven toward an attacker-chosen model; because secure aggregation conceals client-level updates, server-side anomaly inspection may be unavailable precisely when it is most needed [4]. Consequently, secure aggregation should be evaluated jointly with integrity controls, norm bounds, verifiable constraints, or privacy-compatible robust aggregation rather than being treated as a poisoning defense.

Adaptive attackers also invalidate static, defense-agnostic benchmarks. They can formulate optimization objectives that preserve benign-looking distances or norms while maximizing damage, exploit the aggregation rule's acceptance region, and shift malicious parameters into coordinates that benign clients update infrequently [5,8,21,52,80]. Table 5 summarizes how these capabilities challenge recurring defense assumptions.

4. Part II: defense taxonomy in robust federated learning

Once the threat surface is specified, the next question is how defenses should be organized. Rather than treating each proposal as an isolated method, this part groups representative defenses into robust aggregation, backdoor defense, privacy-preserving defense, adaptive poisoning defense, secure aggregation, and verifiable aggregation. Throughout this part, we emphasize a common comparison lens: attack model, required

Table 5
Strong adversary capabilities and the defense assumptions they challenge.

Capability	Representative Strategy	Defense Assumption Challenged	Required Evaluation Dimension	Representative Evidence
Colluding or coordinated clients	Share attack objectives, updates, triggers, and timing; submit mutually consistent malicious updates	Malicious updates are independent outliers or cannot form a coherent cluster	Coalition size, coordination channel, compromised-client selection, and synchronized versus staggered participation	[21,52,66]
Model replacement behind secure aggregation	Scale a malicious update to dominate the aggregate while individual contributions remain hidden	The server can inspect, clip, or reject suspicious client updates; confidentiality implies integrity	Attack success with hidden updates, enforced norm bounds, and compatibility between secure and robust aggregation	[4]
Defense-aware optimization	Optimize malicious updates against Krum, coordinate-wise statistics, anomaly scores, or known thresholds	Fixed attacks adequately measure robustness; benign and malicious updates remain geometrically separable	White-, gray-, and black-box attacker knowledge; adaptive versus non-adaptive baselines; transfer across defenses	[5,21,52]
Stealthy targeted poisoning	Preserve clean accuracy and benign-looking update statistics while inducing attacker-chosen behavior	Clean-task accuracy and simple norm or distance tests reveal successful poisoning	Attack success rate, clean accuracy, detectability, false positives under Non-IID data, and trigger generalization	[4,8,66]
Cross-round adaptive persistence	Schedule attacks across rounds or modify infrequently updated parameters to resist benign retraining	Attacks are one-shot, stationary, and removed by subsequent benign updates	Persistence after attack cessation, intermittent participation, attacker timing, and long-horizon recovery	[80]

Table 6
Defense taxonomy in robust federated learning.

Defense Family	Main Goal	Representative Methods	Key Assumptions	Main Trade-off
Robust aggregation	Filter or down-weight malicious updates before aggregation	AFL, Guard-FL, FedMFG, $\Pi_{p2Brofl}$, and CDRFL	Benign-majority structure, geometric separability, non-coordination, or trusted reference signals	Robustness versus clean-data dependence and computation cost
Backdoor defense	Remove targeted malicious behavior while preserving clean accuracy	BadCleaner	Access to small clean or unlabeled reference data and server-side distillation	Attack removal versus added server overhead
Privacy-preserving defense	Reduce information leakage from updates or outputs	MASKCRYPT, KD + LDP framework, and DP/HE/SMPC baselines	Honest protocol execution and acceptable privacy budget or crypto overhead	Privacy versus utility and communication efficiency
Adaptive poisoning defense	Resist attackers that tailor updates to the aggregation rule	FedDet	Layer-wise anomalies remain detectable despite high dimensionality and heterogeneity	Stronger poisoning robustness versus method complexity
Secure and verifiable aggregation	Protect intermediate updates and validate protocol correctness	TAPFed and VERSA	Threshold trust, bounded collusion, or scalable key establishment	Confidentiality/integrity versus inspectability, scalability, and deployment complexity

trust assumptions, compatibility with heterogeneity, privacy cost, and deployment overhead.

Table 6 provides a high-level map of the defense landscape before we discuss individual methods in detail. It emphasizes that different defense families address different slices of the overall robustness problem and therefore should be compared by target threat, trust assumptions, and cost profile rather than by accuracy alone.

4.1. Robust aggregation against Byzantine and untargeted poisoning attacks

Robust aggregation methods are the most direct response to integrity attacks in federated learning because they attempt to filter or down-weight malicious updates before they contaminate the global model. Their main differences lie in whether they rely on clean server-side references, geometric assumptions on update distributions, reputation mechanisms, or secure computation. Byzantine attacks arising in federated learning are mainly categorized into targeted attacks, such as backdoor and label flipping attacks, which aim to induce the model to make incorrect predictions on specific categories. These are distinguished from non-targeted attacks, specifically gradient perturbation attacks, which disrupt gradient integrity to cause overall model failure. Against such threats, existing defense schemes have significant limitations. Statistical outlier detection-based methods, including Krum and Median, are easily bypassed by carefully crafted adaptive attacks due to their reliance on gradient distribution assumptions. A coalition can further reduce separability by submitting mutually consistent malicious updates or by optimizing all compromised updates jointly against the known aggregation rule [5,21,52]. Meanwhile, auxiliary data validation-based schemes like signSGD and

Zeno++ can be effective only when the proportion of malicious clients remains low, but fail once the proportion exceeds half ($\geq 50\%$), struggling to cope with extreme scenarios involving a majority of malicious participants. The robustness gap of these two types of defense methods in high-intensity attack environments highlights the need for novel defense mechanisms [39].

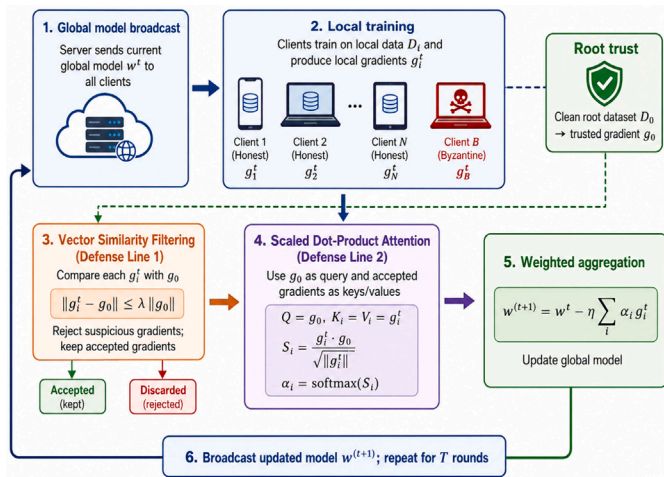
Chen et al. (2023) proposed Attention-based Byzantine-robust Federated Learning with Vector Filter (AFL) to defend against Byzantine attacks in federated learning. AFL combines a vector filter and scaled dot-product attention to build a dual defense, significantly improving robustness and convergence speed. The AFL scheme consists of four stages: local training, vector similarity filtering, scaled dot-product attention based trust scoring, and aggregation. The authors compared AFL with Krum, Bulyan, Fang, Zeno++, and other schemes on MNIST, FMNIST, and CIFAR-10 datasets, evaluating attack robustness and accuracy. Experimental results show that AFL significantly outperforms baseline schemes under four types of attacks (label flipping, Gaussian, gradient derivation, and backdoor attacks) [13]. The results of the ablation experiments are shown in Table 7, together with the default experimental environment reported by the original study. The workflow of AFL is shown in Fig. 6. Moreover, AFL exhibits strong robustness as the malicious-client proportion increases. For example, under a label-flipping attack on FMNIST with 80% malicious clients, AFL maintains approximately 46% average test accuracy, whereas the four evaluated baselines fall below 20%. Additionally, AFL does not increase local computation overhead and requires only a small clean root dataset containing 1000 training samples; its communication overhead is comparable to standard federated learning.

Table 7

Average test accuracy of AFL ablation experiments under the default experimental environment.

Dataset	Scheme	Label Flipping Attack	Gaussian Attack	Gradient Derivation Attack	Backdoor Attack
MNIST	AFL-NoAttention	72.8%	78.9%	80%	69.8%
	AFL-NoFilter	91.5%	84.4%	85.1%	93.9%
	AFL (Full)	92.8%	92.9%	93.1%	94.1%
FMNIST	AFL-NoAttention	74%	75%	79%	59%
	AFL-NoFilter	80%	79%	81%	83%
	AFL (Full)	90%	86%	88%	86%
CIFAR-10	AFL-NoAttention	50%	56.8%	56.3%	57.7%
	AFL-NoFilter	57.3%	59.4%	62.5%	68.4%
	AFL (Full)	64.3%	65.3%	68.4%	69.6%

Experimental environment reported in the original AFL study [13]: 51 selected clients with 10% malicious clients; learning rate $\eta = 0.1$; momentum = 0.9; batch size = 128; and a clean root dataset containing 1000 training samples. MNIST, FMNIST, and CIFAR-10 contain 60,000, 60,000, and 50,000 training instances, respectively, and each contains 10,000 test instances. Experiments were run on Ubuntu 20.04 LTS with an AMD EPYC 7H12 64-Core Processor, 32 GB RAM, and an NVIDIA GeForce RTX 3090 GPU with 24 GB memory. The reported metric is the average accuracy across all test clients. AFL-NoAttention removes the attention mechanism, while AFL-NoFilter removes the vector filter. The original study does not report the model architectures, client-data partitioning, number of local epochs, total communication rounds, or software framework.

**Fig. 6.** Workflow of the attention-based Byzantine-robust federated learning.

However, AFL has several limitations:

- **Root dataset dependency:** It is difficult to guarantee the absolute security of a clean root dataset in practical scenarios. If the root dataset is contaminated, the filtering mechanism may fail.
- **Performance degradation:** Although AFL remains more robust than the evaluated baselines as the malicious-client proportion increases, its average test accuracy still decreases substantially under extreme attack intensity. For example, under a label-flipping attack on FMNIST with 80% malicious clients, AFL achieves approximately 46% accuracy.
- **Privacy gap:** The AFL scheme does not integrate privacy protection techniques, posing a potential leakage risk during client gradient transmission.

Song et al. [54] proposed the Guard-Federated Learning framework to address the problem of Byzantine attacks in federated learning [54]. The core innovation of the Guard-Federated Learning framework lies in effectively tackling the dual challenges of Non-IID data and Byzantine attacks in federated learning through nonlinear dimensionality reduction and generalization capability assessment:

1. **Nonlinear Dimensionality Reduction:** Utilizes uniform manifold approximation and projection to reduce high-dimensional local model parameters into a low-dimensional space. This captures

complex nonlinear relationships to enhance malicious device detection accuracy. On the MNIST dataset, UMAP demonstrated superior cluster separation compared to linear methods like PCA and other techniques like t-SNE. This provides a foundation for subsequent robust regression and K-means clustering, enabling precise identification of malicious devices.

2. **Generalization Capability Assessment:** Introduces the indicative index of weight (IIW) to measure the generalization capability of local models. IIW is approximated using Kullback-Leibler divergence, where a smaller IIW value indicates stronger generalization performance (IIW is positively correlated with the generalization gap). Based on IIW, guard-federated learning employs a dynamic weighted aggregation mechanism. After filtering out malicious devices, it prioritizes aggregating models exhibiting high generalization capability to enhance global model performance.

Experimental validation on the MNIST, Fashion-MNIST, and CIFAR-10 datasets demonstrates the effectiveness of guard-federated learning:

- The UMAP-assisted malicious detection mechanism achieved remarkably high true positive rates of 96–100% against four types of attacks (LF/SF/TRIM/LIE) under extreme non-IID scenarios (Dirichlet concentration parameter $\alpha = 0.1$). Its false positive rate consistently remained $\leq 4\%$, significantly outperforming linear methods like PCA.
- The IIW-driven generalization aggregation mechanism boosted the global model accuracy by an average of 3.9% on CIFAR-10 compared to FedAvg in attack-free scenarios. Against the most stealthy LIE attack, it achieves the lowest attack success rate of 0.45% on MNIST, far below federated learning trust's 2.81%. The strong correlation between the IIW value and generalization capability ensure the rationality of aggregation weight allocation.
- Under extreme conditions, such as a malicious device proportion increased to 40% or heightened data heterogeneity ($\alpha = 0.1$), the model accuracy degradation was less than 4%. This significantly outperformed baseline methods like Median, which suffered performance degradation exceeding 15%.

Sun et al. [58] proposed FedMFG, a reputation-aware framework that models federated learning as a mean-field game in order to improve Byzantine robustness while preserving global learning performance. Its main idea is to estimate a mean-field gradient, compare each client's local gradient against this reference, and then assign reputation scores that

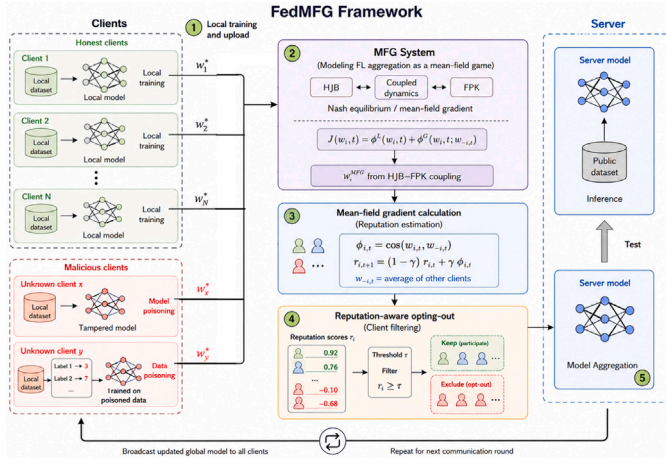


Fig. 7. Representation of FedMFG framework against malicious clients.

support malicious-client exclusion. Experiments on MNIST and Fashion-MNIST show that FedMFG improves both convergence and robustness relative to baselines such as FedAvg, FedProx, Krum, and Multi-Krum, especially under Byzantine attack settings.

FedMFG introduces mean-field game theory into large-scale Byzantine-resilient federated aggregation. It enables dynamic detection and exclusion of malicious clients through a reputation mechanism and provides theoretical analysis of the existence and uniqueness of the mean-field equilibrium and convergence behavior [58]. The workflow of the FedMFG is shown in Fig. 7.

Dong et al. (2024) proposed a secure Top-K protocol, which is based on a shuffle-compare-reveal strategy. This protocol first performs a secure shuffle on the array, then utilizes a divide-and-conquer strategy incorporating quicksort concepts to identify the top-k elements, and finally ensures correctness through batch message authentication code verification.

Building upon this protocol, the authors introduced the $\Pi_{P2Brofl}$ scheme, which is the first federated learning scheme to simultaneously achieve privacy protection and Byzantine robustness. The scheme employs three-party computation to implement a Byzantine-robust aggregation method, protecting the privacy of local models while defending against poisoning attacks [17]. The scheme achieves secure aggregation through the following steps:

- **Scoring Mechanism:** Calculates a score for each participant based on cosine distance to identify potentially malicious models.
- **Secure Selection:** Uses the secure top-k protocol to select the local models with the highest scores for aggregation.
- **Secure Computation:** Adopts a three-party computation protocol to protect privacy during the computation process.

Validation on the Lung Cancer, MNIST, PathMNIST, and CIFAR-10 datasets shows that compared to various aggregation methods, $\Pi_{P2Brofl}$ achieves the best defense performance. When the proportion of malicious participants is 30%, the error rate is 23.16%. When the proportion increases to 50%, the error rate increases by only 1.05%. The scheme performs well across all four different types of datasets. However, the scheme also has certain limitations. It is primarily effective against label-flipping attacks, and its defense capability against other types of attacks (such as backdoor attacks) is limited. Furthermore, integrating other defense methods requires ensuring that the existing security guarantees are not compromised.

Kang et al. [30] designed a lightweight federated learning framework, the CDRFL algorithm, to enhance Byzantine robustness while ensuring user privacy and balancing efficiency with security. The CDRFL algorithm adopts a lightweight chaotic encryption method supporting multiple keys and satisfying ciphertext mutual cancellation, enabling

plaintext recovery during aggregation without decryption. It also employs a dual-server detection mechanism and a reputation scoring mechanism to defend against Byzantine attacks [30].

The CDRFL algorithm excels in privacy protection and Byzantine robustness:

1. **Privacy Protection:** Strong security is achieved through lightweight chaotic encryption; the key space is as large as 2^{4883} under 50 users and 10^{-15} precision, far exceeding the feasible range for brute-force attacks. Pseudo-random sequences based on the Logistic chaotic system ensure ciphertext indistinguishability. Combined with an ideal world/real world model, privacy security is strictly proven under the non-collusion assumption, effectively defending against gradient inference attacks.
2. **Byzantine Robustness:** Under the dual-server non-collusion assumption, a malicious server cannot obtain complete model information; the density-based spatial clustering of applications with noise (DBSCAN) clustering algorithm (parameter $\varepsilon = \text{median of user norms}/2$, $m = \text{number of users}/2$) efficiently isolates anomalous gradients. Combined with a reputation mechanism that dynamically penalizes attackers, model stability is maintained under complex attacks.

Experimental results show that under 30% Byzantine users, CDRFL achieved 98.58%/65.77% accuracy on MNIST/CIFAR-10, outperforming baselines by 3.26%~22.67%. In an industrial scenario, CDRFL accuracy was 3.84% higher than PEMFL. It successfully defended against gradient inference attacks and saved 68.25%~99.63% time compared with the evaluated homomorphic-encryption schemes. Its ciphertext memory requirement was 95.4% lower than AFDL's in the reported comparison. The scalability experiments further show that encryption time increases by 2.85 s as the user count grows from 50 to 200, while communication overhead remains stable from 10 to 200 users because ciphertext size stays close to plaintext size. Nevertheless, CDRFL sends two ciphertexts and therefore communicates more than FedAvg and PEMFL, and its security depends on a dual-server non-total-collusion assumption. The study also does not provide controlled Non-IID, per-round participation-rate, or random-dropout evaluations. The robust federated learning algorithm framework based on CDRFL is shown in Fig. 8.

4.1.1. Cross-method comparison and deployment guidance

A direct ranking of AFL, FedMFG, $\Pi_{P2Brofl}$, and CDRFL by their reported accuracy would be misleading because the original studies use different datasets, models, attack types, attacker ratios, privacy assumptions, and system configurations. We therefore compare their relative strengths through common evidence-oriented dimensions: (i) the tested attack-intensity envelope and attack coverage; (ii) evidence under partial participation, dropout, and changes in the total client population; (iii) evidence under Non-IID data; (iv) communication cost, including cryptographic message expansion and scalability; (v) client- and server-side computation and memory; and (vi) the trust and privacy assumptions required for deployment. Table 8 marks a dimension as directly supported only when it is reported or varied in the corresponding study; otherwise, it is marked as not reported (NR). Importantly, a client-population scalability experiment is not treated as evidence for robustness under a low per-round participation rate or random dropout.

The key assumptions in Table 6 are security boundaries rather than implementation details. Colluding attackers can create a coherent malicious cluster that defeats independent-outlier reasoning; defense-aware attackers can optimize within an acceptance region defined by norms, distances, or validation scores; and secure aggregation can remove the client-level visibility required by anomaly filters. Therefore, a defense claim should state the maximum coalition size, attacker knowledge, visibility of individual updates, adaptation horizon, and whether the guarantee still holds when the adversary knows the defense.

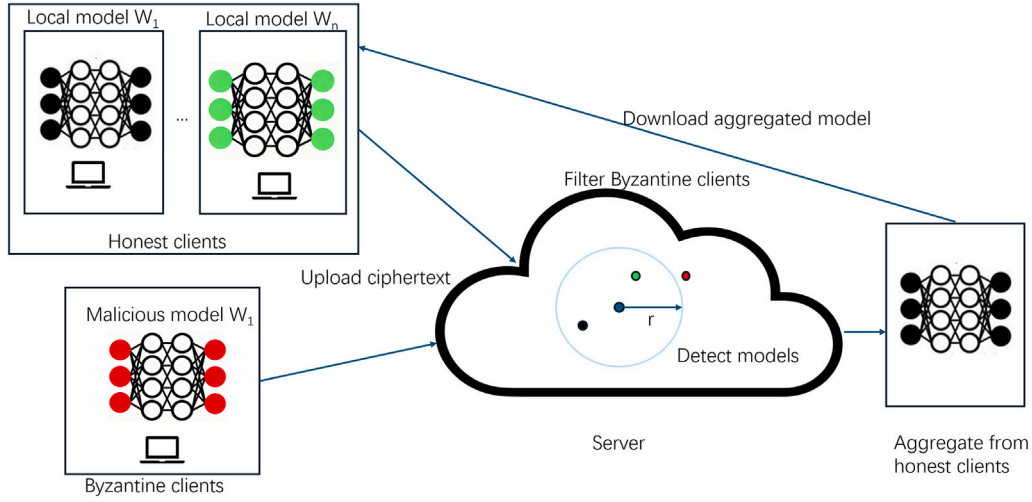


Fig. 8. Robust federated learning framework based on CDRFL.

Table 8
Evidence-oriented comparison of four Byzantine-robust federated learning methods.

Method	Attack Intensity and Coverage	Client Participation	Non-IID Evidence	Communication Budget	Computation / System Assumptions	Best-Supported Deployment Fit
AFL [13]	Direct: 10%–80% malicious clients under label-flipping, Gaussian, gradient-derivation, and backdoor attacks. At 80% label-flipping attackers on FMNIST, AFL retains about 46% accuracy.	Fixed set of 51 chosen clients; participation-rate, dropout, and population-scale sweeps are NR .	NR: client-data partitioning is not reported.	No secure-computation layer; quantitative bytes, latency, and rounds-to-target are NR .	Claimed no added local computation; server filtering and attention require a trusted 1000-sample root dataset.	Broad attack coverage and high attacker ratios when trusted root data are feasible and update confidentiality is not required.
FedMFG [58]	Direct: four attackers among 80 clients for model/data poisoning; also varies poisoned-data level from 10% to 40% under IID and Non-IID data. High malicious-client ratios are NR .	Tests populations from 10 to 300 clients and models opting out; reports that 50 clients approximate the tested large-scale setting. Random participation/dropout sweeps are NR .	Direct: controlled IID versus Non-IID experiments on MNIST and FMNIST.	Quantitative bytes, latency, and message expansion are NR .	Mean-field calculation and reputation add server computation; smaller γ improves accuracy but takes longer. Runtime/memory are NR ; assumes approximately homogeneous, rational clients.	Large-scale Non-IID IoT learning with relatively low attacker prevalence and no requirement for built-in update confidentiality.
Π_{P2Brot} [17]	Direct: 10%–50% malicious participants; CIFAR-10 error rises only from 22.93% to 23.98% at 50%. Evaluation is limited to label flipping.	Uses 100 participants for robustness and varies the population from 200 to 500 for scalability. Participation-rate and dropout sweeps are NR .	Direct limitation: experiments use IID data with equal samples per participant.	Direct: three-server LAN measurements; at fixed population, overhead is insensitive to attacker ratio, while communication and runtime grow linearly with total participants.	Invokes secure Top-K $n + 1$ times per round; assumes three computation servers with at most one malicious server.	Privacy-critical label-flipping settings when three-party infrastructure and linearly growing secure-computation cost are acceptable.
CDRFL [30]	Direct: 5%–30% Byzantine users under free-rider and label-flipping attacks, plus gradient-inference evaluation. Evidence above 30% is NR .	Defaults to 20 users; evaluates computation at 50–200 users and communication at 10–200 users. Participation-rate and dropout sweeps are NR .	NR: no controlled IID/Non-IID comparison is reported.	Two ciphertext uploads make cost higher than FedAvg/PEMFL, but ciphertext size is near plaintext; reported memory is 95.4% below AFDL and communication remains stable from 10 to 200 users.	Adds 0.37 s over FedAvg in the reported setting and is 68.25%–99.63% faster than selected HE baselines; relies on dual servers and a non-total-collusion assumption.	Privacy-aware edge or industrial systems seeking quantified efficiency and scalability evidence within the tested attack range.

Legend: Direct: the corresponding factor is reported or varied in the original study; NR: not reported as a controlled evaluation dimension. Reported utility values should not be directly ranked across rows because the experimental settings differ. Deployment-fit statements combine direct evidence with explicit mechanism-based implications.

Evaluations that vary only the malicious-client percentage but keep attackers independent and attack-agnostic may substantially overestimate deployment robustness [4,5,21,52].

Several practical selection rules follow from the full-text evidence. When attack-type coverage and malicious-client intensity dominate the decision, AFL provides the broadest stress test, spanning four targeted and untargeted attacks and attacker ratios up to 80%; however, it relies

on trusted root data, exposes updates, and provides no controlled Non-IID or resource-budget evaluation. FedMFG is the only one of the four with direct controlled IID/Non-IID comparisons and the broadest client-population study, ranging from 10 to 300 clients. It is therefore the best-supported choice for large-scale heterogeneous IoT learning, but its Byzantine experiment contains only four attackers among 80 clients, and it does not quantify communication or computation costs.

When update confidentiality must be integrated with Byzantine defense, the choice is between different trust and budget profiles. $\Pi_{P2Brofl}$ offers the strongest explicit malicious-security model, including privacy and correctness with three computation servers and evidence at 50% malicious participation, but its evidence is limited to IID label-flipping experiments and its communication and runtime grow linearly with the total participants. CDRFL covers two Byzantine attacks and gradient inference, reports only 0.37 s additional runtime over FedAvg in its setting, and provides direct scalability measurements; nevertheless, it is evaluated only up to 30% Byzantine users, sends two ciphertexts, and requires its dual-server non-total-collusion assumption to hold.

Under changing client availability, FedMFG has the strongest mechanism-level fit because its opting-out and reputation mechanisms explicitly account for stragglers and low-contribution clients, and its experiments cover the largest population range. AFL is comparatively attractive when a trusted root dataset is available because each round's updates can be assessed against a server-derived reference without requiring a multi-server cryptographic protocol; however, its fixed 51-client evaluation does not establish behavior under sparse participation. CDRFL provides stronger measured scalability evidence than AFL and FedMFG, but intermittent identities may weaken its reputation history. $\Pi_{P2Brofl}$ provides the strongest privacy guarantee, yet its coordinated three-party secure computation and linearly increasing cost make it the least lightweight option when participation is highly dynamic. None of the four studies directly varies per-round participation rate or random dropout, so these relative advantages follow from their mechanisms and reported population-scale evidence rather than a controlled head-to-head experiment.

4.2. Defense methods against backdoor attacks

Backdoor defenses differ from generic Byzantine-robust aggregation because they must suppress targeted malicious behavior while preserving clean-task accuracy. This often creates a three-way tension among attack removal, clean-data requirements, and computational cost. Federated learning, as a privacy-preserving distributed learning paradigm, allows multiple participants to train models locally and share only model parameters instead of raw data. However, federated learning systems have critical security vulnerabilities, particularly backdoor attacks. Malicious participants embed triggers such as specific pixel blocks, watermarks, or noise into local training data, causing the global model to misclassify inputs containing the trigger into a target label. This behavior is exemplified by the misclassification of watermarked images as 'stop signs'.

The two main types of backdoor attacks in federated learning are:

- **Model Replacement Attack:** Attackers train an attacker-chosen local model and scale its update to offset aggregation dilution, thereby driving the next global model toward the malicious model. The attack can remain effective when secure aggregation hides individual updates, because confidentiality prevents client-level inspection but does not constrain the integrity of the aggregate [4].
- **Distributed or Coordinated Attack:** Multiple compromised clients collaborate by decomposing a global trigger into local trigger patterns and coordinating their poisoning behavior. DBA shows that the composed trigger can produce a global backdoor while each local contribution appears less suspicious than a single concentrated attack [66].

Stealth is therefore multi-dimensional: an attack may preserve clean-task accuracy, remain within norm or distance thresholds, distribute its effect across clients, and persist after compromised clients stop participating. Constrain-and-scale optimization explicitly trades attack success against anomaly detectability, while Neurotoxin targets parameters that benign clients update less frequently to prolong backdoor persistence [4,80]. Backdoor defenses should consequently report coalition size,

Table 9
Privacy attack types.

References	Attack Type	Attack Target	Key Exposure or Attack Condition
[35,39]	Membership Inference Attack (MIA)	Determine whether a specific record participated in training.	Requires at least query access; overfitting and poor generalization increase exposure.
[35]	Model Extraction Attack (MEA)	Reconstruct an approximate or exact copy of the target model.	Repeated black-box queries and access to model outputs can train a substitute model.
[35,39]	Property Inference Attack (PIA)	Infer latent properties or characteristics of the training data.	May exploit model updates, auxiliary data, or shadow classifiers in white-box or black-box settings.
[35,39]	Data Reconstruction Attack (DRA)	Reconstruct training inputs and labels from gradients or model updates.	Requires access to gradients or updates; gradient-matching attacks may recover samples within few iterations.

attacker timing, update visibility, clean accuracy, attack success rate, detectability, and persistence after attack cessation.

However, many backdoor defenses, such as pruning, fine-tuning, and anomaly detection, depend on labeled clean data and often degrade clean-task accuracy under Non-IID conditions. BadCleaner addresses this limitation through attention-based multi-teacher distillation with a small amount of unlabeled clean data, reducing attack success rates to below 3% in the reported settings while largely preserving task accuracy [78]. The trade-off is additional server-side distillation cost, which makes lightweight backdoor removal an important direction for practical deployment.

4.3. Privacy-preserving defense mechanisms

Privacy-preserving defenses target confidentiality threats such as membership inference, property inference, and gradient inversion. In practice, their main design trade-off is whether privacy is obtained through stronger cryptography, stronger noise injection, or more communication-efficient knowledge transfer (Table 9).

Although federated learning protects privacy through local data training and model weight exchange, the exposed weights are vulnerable to attacks such as membership inference and gradient inversion attacks. Existing methods like homomorphic encryption provide strong security guarantees but cause drastic communication overhead (model size inflates 20-200x after encryption), while differential privacy compromises model accuracy [22].

To address this, Hu et al. (2025) proposed MASKCRYPT, a selective homomorphic encryption framework for federated learning that explicitly targets the trade-off between privacy protection and system overhead. The core idea of MASKCRYPT is encryption sparsity, meaning that only a small portion of critical model weights, such as 5–25%, is encrypted rather than the entire model [27].

Experimental results show that MASKCRYPT achieves an effective trade-off between privacy protection and system overhead in federated learning through selective homomorphic encryption. Its two main components, gradient-guided mask selection and the mask consensus mechanism, enable defense against membership inference and data reconstruction attacks with partial weight encryption ($p \leq 20\%$), while reducing communication overhead by 4.15x. Its effectiveness is also validated on diverse model families, including CNNs and Transformers.

Despite significant achievements of MASKCRYPT in balancing privacy and efficiency, its design still has limitations: First, the mask consensus mechanism, as a critical step, necessitates passing mask proposals and the final global mask (typically represented as a bitmap)

between the clients and the server, inevitably introducing additional communication overhead. Although using bitmaps can mitigate this overhead, it remains an additional cost compared to the base protocol without masking. Second, the security of MASKCRYPT heavily relies on client participation in decryption collaboration (using their private key shares). In dynamic client scenarios, if some clients disconnect unexpectedly after aggregation but before decryption is complete, the server will be unable to decrypt the aggregated ciphertext model updates, causing the entire round to fail and blocking the training process.

Gad et al. (2023) combined knowledge distillation with local differential privacy to reduce communication while supporting heterogeneous client models [24]. Instead of exchanging full model parameters, clients upload soft labels on a public dataset, the server aggregates these labels, and clients then refine local models through distillation. This design is attractive because it naturally lowers communication costs and preserves architectural flexibility across clients. Reported results on multimodal HAR datasets nevertheless show a clear utility–privacy tension: under stronger privacy budgets, performance degrades substantially, especially for high-dimensional image data, whereas simpler tabular modalities retain better utility. The broader implication is that distillation-based privacy mechanisms can improve deployability, but their effectiveness remains highly data dependent.

4.4. Defense mechanisms against adaptive model poisoning

Adaptive poisoning defenses are particularly important because strong attackers can tailor malicious updates to the aggregation rule itself. Compared with generic robust aggregation, these methods place greater emphasis on attacker knowledge, layer-wise behavior, and robustness under high-dimensional optimization. Adaptivity should be interpreted broadly. A defense-aware adversary may know the exact aggregation rule and thresholds, estimate benign updates using compromised clients, optimize a malicious update within the defense's acceptance region, coordinate several malicious submissions, and revise its strategy after observing later global models. Early stealthy model-poisoning work formulated alternating minimization objectives that balance targeted damage and evasion, while subsequent attacks explicitly optimized against Byzantine-robust aggregation rules or hid within the natural variance of benign updates [5,8,21,52]. Robustness against a fixed library of non-adaptive attacks therefore does not imply robustness against an attacker who knows the deployed defense. Recent privacy-preserving work further shows that model-poisoning defense cannot simply assume plaintext inspection. The robust privacy-preserving model of Yazdinejad et al. uses additive homomorphic encryption to protect local gradients while introducing an internal auditor that analyzes encrypted-gradient similarity and distribution through Gaussian mixture modeling and Mahalanobis distance [75]. This line of work is important because encrypted gradients can hide both private information and malicious behavior; therefore, robustness claims should report whether the defense works before or after decryption, what leakage the auditor requires, and how computational overhead changes under IID and Non-IID client data. Federated learning enables privacy-preserving distributed training in Internet of Things scenarios but is vulnerable to model poisoning attacks, such as backdoor attacks and model accuracy disruption attacks. Adaptive model poisoning attacks are difficult to detect and can bypass existing defense mechanisms, including Krum and Median, by customizing malicious parameters to destroy model usability. Following the unified reformulation by Yang et al., Table 10 summarizes four representative adaptive attack families [72].

Existing defense mechanisms are fragile against such attacks, primarily for two reasons: the high dimensionality of model parameters and data heterogeneity. High parameter dimensionality allows attackers to optimize only a subset of parameters, such as those infrequently updated, thereby bypassing full-parameter distance detection methods like Krum. Furthermore, parameter-level aggregation methods including Median and Trimmed-Mean deviate from the optimal

Table 10

Reformulated adaptive model-poisoning attacks and their characteristics.

References	Attack Type	Optimization Objective	Core Constraint
[21,72]	STAT-OPT	Maximize global-model direction deviation $s^T(G - G')$.	Both G and G' are produced by the target aggregator; the first m client updates are replaced by optimized malicious updates.
[52,72]	DYN-OPT	Maximize the L2 distance between benign and compromised global models, $\ G - G'\ _2$.	Each malicious update is restricted to $w'_i = G + \gamma \nabla^p$, and G' is produced by the target aggregator.
[52,72]	Min-Max	Maximize the distance $\ w' - w_i\ _2$ between a malicious update and benign updates.	The malicious update follows $w' = G + \gamma \nabla^p$, and its distance to a benign update is bounded by the maximum pairwise benign-update distance.
[52,72]	Min-Sum	Maximize the sum of squared distances $\sum_i \ w' - w_i\ _2^2$ from the malicious update to benign updates.	The malicious update follows $w' = G + \gamma \nabla^p$, and its summed squared distance is bounded by the maximum summed pairwise benign-update distance.

model by excluding potentially malicious parameters. To address this, Yang et al. (2024) proposed a novel robust federated learning algorithm named FedDet. The core innovation of FedDet lies in splitting local models by layer before aggregation and processing them with median normalization, effectively tackling the challenges of high dimensionality and data heterogeneity [72].

In experimental validation, using the FEMNIST and CIFAR-10 datasets with CNN (for FEMNIST) and AlexNet (for CIFAR-10) model architectures, and setting malicious client proportions from 5% to 30%, FedDet was compared with baseline methods like Krum, Multi-Krum, Median, and Trimmed-Mean. The study reformulates four existing attack families—STAT-OPT, DYN-OPT, Min-Max, and Min-Sum—and evaluates six adaptive attack variants tailored to FedDet. The results show that the accuracy of FedDet only dropped from 60.43% to 57.11% under a 20% malicious proportion STAT-OPT attack. This significantly outperformed baselines; for instance, under a 20% malicious proportion DYN-OPT attack, FedDet accuracy reached 41% while Krum dropped to 30%. This verifies the effectiveness of the layer splitting of FedDet and median normalization. Experiments also analyzed the impact of attacker knowledge levels, finding that FedDet had higher accuracy against weak attackers, achieving 30% versus 15% under a strong attack at a 30% malicious proportion. However, the current work does not cover targeted model poisoning attacks, particularly backdoor attacks, which can insert stealthy triggers while maintaining overall model accuracy, causing misclassification of specific samples. This leaves security risks in privacy-sensitive scenarios, such as medical IoT.

Future research will focus on combining the layer-aware mechanism of FedDet with other defense techniques, including gradient-based anomaly detection, to build a unified framework for defending against both targeted and non-targeted attacks. We will also extend this to practical deployment in multi-device collaborative learning, for example in unmanned aerial vehicle (UAV) networks, and in dynamic attack scenarios.

4.5. Secure aggregation beyond honest-but-curious assumptions

Secure aggregation is often introduced as a privacy primitive, but in robust federated learning it must also be examined as a system design choice. Once stronger adversaries, collusion, and aggregator failures are considered, privacy, scalability, and reliability become tightly coupled. Importantly, secure aggregation protects update confidentiality rather than update correctness. If a protocol reveals only an aggregate, the server may be unable to run client-level anomaly detection, attribution,

or rejection. Model replacement demonstrates how a malicious client can exploit this loss of inspectability by scaling its update to steer the aggregate toward an attacker-chosen backdoored model [4]. Colluding clients can distribute this influence across several hidden contributions, further weakening per-client defenses. Secure aggregation protocols should therefore state their integrity guarantees separately from their privacy guarantees and should be evaluated with privacy-compatible clipping, validity proofs, verifiable constraints, or Byzantine-robust secure computation. The core privacy problem in federated learning stems from honest-but-curious (HbC) aggregator behavior during gradient exchange. These aggregators, while adhering to protocol rules, might infer private information of participants by accessing intermediate aggregated models, such as global gradients from training rounds. This access can lead to gradient inversion attacks (reconstructing raw data from gradients) or decomposition attacks (inferring updates of individual participants through multiple rounds of aggregation). This vulnerability exposes a severe limitation of existing secure aggregation schemes. Mainstream methods, like those based on homomorphic encryption, functional encryption, or masking techniques, mostly rely on a single HbC aggregator architecture, making systems vulnerable to attacks and prone to single points of failure, while also incurring high communication costs. Specifically, the single-aggregator architecture cannot defend against intermediate model leakage. Alternative multi-aggregator solutions, including secure multi-party computation or trusted execution environments, attempt to decentralize trust but require full peer-to-peer communication. This not only suffers from poor scalability, causing latency bottlenecks in large networks, but is also vulnerable to aggregator failures, where the failure of one node causes the entire aggregation to crash. Ultimately, this makes it difficult to balance privacy, efficiency, and reliability.

Xu et al. (2024) summarized the limitations of existing methods, highlighting that HbC aggregator architectures such as “Single HbC A +” are vulnerable to isolation attacks, replay attacks, and gradient inference attacks. While these attacks do not necessarily crash the system, they can render privacy protection ineffective and violate the core goal of federated learning. Therefore, they proposed TAPFed, a threshold functional encryption-based framework with a multi-aggregator architecture that can tolerate malicious aggregators while significantly reducing communication overhead. Furthermore, TAPFed aims to defend against several known attack vectors, including gradient inference, decomposition, isolation, replay, and collusion attacks. Experimental results demonstrate the efficiency of TAPFed on MNIST and CIFAR-10 across multiple dimensions: model accuracy remains comparable to unencrypted federated learning, training time is reduced relative to several HE-based baselines, and communication overhead is reduced by 29%–45% in the reported settings [71].

Although TAPFed innovatively addresses privacy problems of federated learning, it still has core limitations. Firstly, TAPFed is explicitly designed for cross-silo federated learning, involving a small number of organizations, such as hospitals and banks, which have relatively sufficient computing resources. It cannot be directly applied to cross-device scenarios, including large-scale IoT devices involving tens of millions of resource-constrained devices, as the threshold functional encryption of TAPFed requires certain computational capabilities, specifically integer group operations, which may be unfriendly to low-power devices. Secondly, experimental results show that communication overhead increases linearly with the number of aggregators. While increasing the number of aggregators enhances fault tolerance, it significantly increases transmission load. Thirdly, the TAPFed framework does not address security attacks, such as backdoor or poisoning attacks, focusing solely on privacy leakage risks. Additionally, experimental validation has limitations: performance was only evaluated on MNIST/CIFAR-10 (image classification) using a CNN model, without covering complex tasks like text or speech, or large models like Transformers.

4.6. Verifiable aggregation against model recovery attacks

Verifiable aggregation complements secure aggregation by asking whether participants and servers can efficiently confirm that aggregation was performed correctly without exposing sensitive local models. This line of work is especially relevant when robustness depends not only on secrecy but also on trust in the protocol execution itself. In traditional federated learning, users locally train model updates, encrypt them, and upload them; the server aggregates updates into the global model. A security vulnerability exists: shared keys can lead to model recovery attacks, where attackers can brute-force hashes to leak up to 98% of the local model. Existing schemes like VerifyNet rely on pre-shared keys among users, which are difficult to implement for trusted initialization in cross-device scenarios (resource-constrained, unreliable networks). Furthermore, VerifyNet uses heavy operations like bilinear pairings, making verification inefficient for high-dimensional models.

Hahn et al. (2023) proposed the VERSA protocol. VERSA reuses the pairwise key agreement from Secure Aggregation (SA), eliminating the need for pre-shared keys. Moreover, it implements verification based on a Pseudo-Random Generator (PRG), avoiding bilinear pairings. Experimental results show VERSA verification is 5 orders of magnitude faster than VerifyNet (15 ms vs. 20 min), adding only 2x overhead to SA. It maintains accuracy on MNIST/SVHN/CIFAR-100 datasets and supports high-dimensional models [26].

While VERSA achieves significant progress in cross-device federated learning, it still has limitations. Firstly, the protocol does not cover poisoning attacks and membership inference attacks. The former allows malicious users to manipulate the global model by distorting local model updates, such as submitting parameters outside the expected range. Meanwhile, the latter could potentially infer training data privacy, including the presence of specific data items, from aggregation results. These attacks are increasingly prominent in federated learning but are not addressed by the current design. Secondly, in ultra-large-scale user scenarios (number of users $> 10^4$), the overhead of key sharing and masking communication could become a performance bottleneck, requiring optimization of the communication protocol to reduce bandwidth costs.

5. Part III: cross-cutting trade-offs in robust federated learning

Many important challenges in robust federated learning do not belong to a single attack or defense family. Instead, they arise from tensions between robustness and other objectives, such as noisy supervision, out-of-distribution generalization, fairness, communication efficiency, and client personalization. This part therefore focuses on cross-cutting trade-offs. The common theme is that robustness must be evaluated jointly with utility, adaptability, and cost rather than as an isolated metric.

Table 11 summarizes the main trade-offs discussed in this part. The goal is not to assign absolute scores, but to highlight where existing methods repeatedly face structural tensions between robustness and another design objective.

5.1. Robustness under noisy labels

Noisy labels illustrate how robustness can degrade even without a fully malicious adversary. In realistic deployments, mislabeled data, weak supervision, and partial corruption often interact with Non-IID effects and complicate the interpretation of model quality. Federated learning is a distributed machine learning paradigm designed to leverage data from edge nodes while protecting privacy. However, edge nodes often collect data with noisy labels, such as those resulting from malicious attacks or labeling errors, leading to degraded model performance. Traditional federated learning ignores the data noise problem, using a

Table 11
Cross-cutting trade-off matrix in robust federated learning.

Trade-off Pair	Why the Tension Arises	Representative Discussion in This Survey	Typical Failure Mode	Representative Sources
Robustness vs. privacy	Strong privacy mechanisms hide useful signals for anomaly detection or inject noise into training	MASKCRYPT, KD + LDP, TAPFed, and VERSA	Better privacy but weaker utility or higher protocol cost	[22,26,71]
Robustness vs. heterogeneity	Non-IID data makes benign and malicious updates harder to separate	Guard-FL, FedDet, DSPFL, and Sylva	False positives, weaker convergence, and unstable personalization	[46,54,70,72]
Robustness vs. communication efficiency	Compression and infrequent communication distort gradients and delay correction	STC, signSGD, FedAvg, and dataset distillation	Faster training but less stable learning under skew	[38,50,61]
Robustness vs. fairness/personalization	Defenses optimized for the average client may fail on minority groups or client-specific distributions	FairGuard, Sylva, and DSPFL	Good global accuracy but poor subgroup or local robustness	[46,53,70]

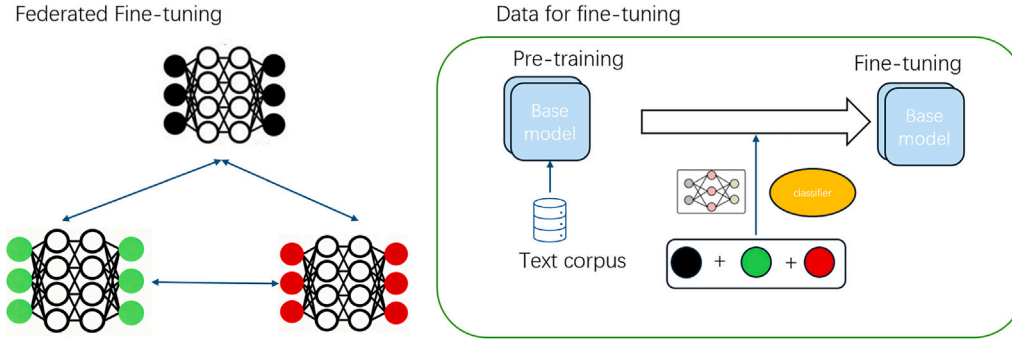


Fig. 9. Deployment Pipeline of Federated Fine-Tuning.

uniformly weighted loss function, making it difficult to converge to a high-quality model in noisy environments.

Chen et al. [14] proposed a reweighted federated learning framework to address noisy labels through two variants: D-DAD for settings with complete clean-data support and D-ADMM for partial-data settings. Compared with conventional federated learning, which uniformly weights local losses, these methods explicitly reweight noisy supervision and thereby improve robustness to corrupted labels. Reported experiments show clear gains on MNIST and CIFAR-10, especially when the noise ratio is below 50%. However, the approach still assumes relatively reliable communication, degrades under extremely high noise ratios, and requires further study in more general Non-IID environments.

5.2. Robustness of federated fine-tuning for pretrained models

Pretrained models introduce a different trade-off surface: federated learning must preserve the transferability and robustness of a strong foundation model while adapting to highly skewed local data and limited communication budgets. Federated learning addresses data scarcity and privacy concerns through distributed training. However, during the federated fine-tuning phase, communication constraints and data heterogeneity characterized by skewed label and quantity distributions become apparent. These factors can distort features and degrade model performance on out-of-distribution data, compromising the robustness of the pre-trained model, defined as its stability under data distribution shifts. This impacts critical applications like autonomous driving and medical diagnosis. Du et al. (2025) experimentally discovered that federated learning distorts the feature space, causing significant performance drops on out-of-distribution datasets. The deployment pipeline of federated fine-tuning is shown in Fig. 9.

They proposed three robustness metrics:

1. Singular Value Entropy: Measures the complexity of the feature space (higher singular value entropy indicates stronger representational capacity).

2. Largest Singular Value Ratio: Assesses model transferability (lower largest singular value ratio indicates a more uniform feature distribution).
3. Gradient Deviation Angle: Quantifies the deviation angle between the model and the pre-trained model (lower gradient deviation angle indicates better retained robustness).

They also proposed the Gaussian Noise Projection algorithm (GNP). GNP extracts robust vectors from the fine-tuned model via projection operations, inheriting the robustness of pre-trained model; Gaussian noise is added to enhance representation capability [18].

The RoBERTa-Base model was tested on sentiment analysis datasets (Amazon-ID; DynaSent, SemEval, and SST-3-OOD). Federated learning simulated label skew ($\alpha \in \{10.0, 1.0, 0.1\}$) and quantity skew ($\beta \in \{0.1, 0.3, 0.6\}$). Experimental results show that under label skew, GNP significantly boosts out-of-distribution accuracy. Specifically, at $\alpha = 0.1$, SST-3 accuracy increased from 37.4% to 50.3% for Prefix Tuning and from 37.4% to 67.3% for Adapter. Under quantity skew, GNP stabilizes metrics and outperforms baselines such as FedAvg and FedIIR, evidenced by achieving 72.6% SST-3 accuracy at $\beta = 0.6$. The GNP algorithm effectively controls singular value entropy decline and gradient deviation angle fluctuations. Furthermore, it demonstrates strong generality, as it is applicable to various parameter-efficient fine-tuning methods and does not rely on prior knowledge of client data. However, the algorithm faces a challenge regarding hardware heterogeneity by not accounting for differences like the computational limits of mobile devices in cross-device federated learning. There is also insufficient dataset coverage, as experiments focused solely on text sentiment analysis tasks, leaving the robustness impact on multimodal data including images and speech unverified.

5.3. Fairness under adversarial and Non-IID conditions

Fairness is included here because it cuts across both attack and defense taxonomies. A method can be robust in the usual accuracy sense but still fail under subgroup-specific manipulation, making fairness a

distinct robustness criterion in adversarial federated learning. Algorithmic fairness in federated learning faces exacerbated challenges due to the Non-IID data (where significant discrepancies may exist between individual clients' data distributions and the global distribution across all clients), as well as its heightened vulnerability to adversarial attacks. Furthermore, given the privacy constraints in federated learning (where servers cannot directly access client data), fairness assurance algorithms designed for centralized machine learning environments (which typically rely on centralized access to sensitive attributes in the training set) cannot be directly applied to the federated learning setting.

Traditional research on fairness in federated learning has primarily focused on biases inherent in the training data or introduced by Non-IID data, operating under the assumption of trusted and cooperative participants. However, this assumption no longer holds in adversarial environments, posing serious threats to model fairness.

Sheng et al. [53] introduced four types of fairness attacks based on partitioning data samples by sensitive attributes and labels: label-flipping fairness attacks, attribute-flipping fairness attacks, hybrid-flipping fairness attacks, and dual-flipping fairness attacks. These attacks compromise the fairness of federated learning models by manipulating sensitive attributes and labels in the training data. Existing fairness-enhancing algorithms are insufficient to mitigate the proposed attacks. To address this issue, the authors proposed FairGuard, a defense framework whose main idea is a collaborative mechanism that combines server-side dual-inference detection with client-local debiasing. Malicious clients are identified and excluded during global aggregation to improve fairness under adversarial conditions.

Experiments indicate that FairGuard performs well across multiple settings. On the Adult [19] and COMPAS [36] datasets, fairness metrics such as average equalized odds difference and average statistical parity difference approach the corresponding no-attack baselines. The method also maintains relatively stable performance under different Non-IID data distributions and varying attack intensities.

5.4. Robustness–communication efficiency trade-offs

Communication efficiency is another cross-cutting issue because almost every robust federated learning method ultimately depends on repeated update exchanges. Compression can improve deployability, but it may also distort gradients and reduce robustness under heterogeneity. Federated learning allows multiple clients to collaboratively train a model without sharing local data, but repeated exchanges of high-dimensional model updates create substantial communication overhead. Existing methods reduce this overhead either by decreasing the number of exchanges or by compressing the messages sent in each exchange [43,50]. Eq. (2) summarizes these factors:

$$b^{\text{up/down}} \in \mathcal{O}\left(N_{\text{iter}} \times f \times |W| \left(H(\Delta W^{\text{up/down}}) + \eta\right)\right) \quad (2)$$

where N_{iter} is the number of training iterations, f is the communication frequency, $|W|$ is the model size, H is the update entropy, and η is the encoding efficiency loss. The three representative baselines below illustrate

different ways to reduce these factors, but none simultaneously guarantees symmetric message compression and stable convergence under severe client heterogeneity.

Key problems with existing methods are:

- **Federated Averaging (FedAvg)**: Performs several local optimization steps before full-model exchange, thereby reducing the number of communication rounds rather than compressing each transmitted model. Its original evaluation demonstrated substantial reductions in communication rounds, while later federated-learning comparisons found that convergence can deteriorate under severe Non-IID data, small local batches, or low client participation [43,50].
- **signSGD with majority vote**: Quantizes every gradient coordinate to one bit and aggregates worker signs by majority vote. The original method reports a 32× per-iteration communication reduction relative to 32-bit gradients and supports sign communication in both directions, but federated-learning experiments show sensitivity to low gradient-sign congruence under severe Non-IID partitioning [7,50].
- **Top- k sparsification with error feedback**: Transmits the k largest-magnitude coordinates and retains the omitted information in memory for later updates. This mechanism can strongly reduce uplink traffic while preserving convergence under the assumptions analyzed for sparsified SGD, but conventional federated use does not directly yield an equally sparse downlink because client-specific sparse supports are not preserved after aggregation [50,55].

Table 12 separates the original communication-saving mechanism of each method from the federated-learning evidence concerning directionality and Non-IID behavior.

Sattler et al. proposed the sparse ternary compression (STC) framework, which combines ternarization, residual accumulation, and Golomb encoding to improve communication efficiency while remaining relatively robust under Non-IID data [50]. Compared with FedAvg and signSGD, STC performs better in highly heterogeneous settings and under limited client participation, while also substantially reducing communication costs. Its limitations are that accuracy can drop under extreme compression, encoding introduces extra latency, and the method does not natively address privacy leakage.

Additionally, dataset distillation as a data compression technique can also alleviate communication overhead issues. Dataset distillation synthesizes a small set of highly informative data (synthetic set S) to replace the original large-scale dataset T , enabling models trained on S to achieve generalization performance comparable to training on T .

Lei and Tao [38] categorized dataset distillation into two broad frameworks: meta-learning methods and data-matching methods. Among them, factorized dataset distillation is particularly notable because it decomposes synthetic data into an encoder and a generator, thereby improving compression efficiency. Existing results suggest that factorized methods perform strongly on several benchmarks, while

Table 12

Evidence-linked comparison of representative communication optimization methods in federated learning.

Method and Sources	Communication-Saving Mechanism	Direction Affected	Non-IID Behavior in Federated Learning
signSGD with majority vote [7,50]	One-bit sign quantization; 32× per coordinate relative to 32-bit gradients	Uplink and downlink	Sensitive to severe client-level class skew
Top- k sparsification with error feedback [50,55]	Transmit the k largest-magnitude coordinates; reduction depends on k/d and index-encoding cost	Primarily uplink in conventional FL	Relatively robust in the cited FL comparison, but aggregated downlink sparsity is not preserved
Federated Averaging (FedAvg) [43,50]	Multiple local steps between full-model exchanges; saves rounds rather than bits per update	Fewer exchanges in both directions	Sensitive to severe skew, small batches, and low participation

Original method sources establish the communication mechanisms; the directionality and Non-IID assessments follow the federated comparison of Sattler et al. [50]. “Relatively robust” denotes comparative empirical behavior rather than a formal robustness guarantee.

Table 13

Representative performance of dataset distillation methods on standard datasets at images per class (IPC) = 50.

Dataset	Best Method (IPC = 50)	Accuracy	Comparison (Full Dataset)
CIFAR-10	KFS (Factorized) [37]	92.2%	93.5%
CIFAR-100	KFS [37]	75.0%	84.8%
Tiny ImageNet	KFS [37]	40.0%	37.6%
ImageNet-1K	TESLA (TM) [15]	27.9%	33.8%

kernel-ridge-regression and trajectory-matching methods remain competitive among non-factorized approaches. Table 13 reports representative IPC = 50 results from the original KFS and TESLA studies [15,37].

Although factorized methods are optimal for most tasks, the generated high-resolution images exhibit aliasing artifacts. Furthermore, synthetic data is prone to overfitting the architecture used during distillation, leading to significant performance degradation when transferred to ResNet (average drop >10%). Factorized methods demonstrate superior transferability.

5.5. Personalization under heterogeneity and adversarial risk

Personalization becomes necessary when a single global model cannot adequately serve all clients. However, personalized training also changes the attack surface and the meaning of robustness, since defenses must now protect both shared and client-specific components. In multi-client environments, federated adversarial training has proven to be a powerful method for enhancing model robustness while preserving privacy. With the rise of large pre-trained models, fine-tuning via FAT is expected to become mainstream, enabling devices to contribute to their limited data through periodic synchronization and knowledge sharing. Despite its potential, most existing research focuses on building generalized global models to address Non-IID data challenges. However, under significant data heterogeneity, such global models often fail to provide effective defenses tailored to individual clients. While their strong generalization capability is beneficial in some cases, it limits their ability to adapt to unique client-specific data distributions, leading to performance degradation. Furthermore, integrating federated adversarial training with large pre-trained models introduces communication overhead that can significantly impact training speed, especially in resource-constrained environments [57].

To address this issue, Qi et al. [46] proposed Sylva, a personalized defense architecture for collaboratively fine-tuning pre-trained models. Sylva aims to improve both benign accuracy and adversarial robustness under heterogeneous client distributions while keeping communication overhead manageable. It uses low-rank adaptation (LoRA) so that only 0.7% of model parameters need to be transmitted, and the reported communication size for ViT-B is as low as 2.06MB. The framework adopts a two-stage collaborative optimization mechanism: Stage 1 performs federated LoRA fine-tuning with ball-tree aggregation to build a shared defense backbone, and Stage 2 applies Shapley value-based layer selection to unfreeze a small set of important layers for personalization.

The framework is also evaluated on edge devices and shows that lightweight deployment is feasible in resource-constrained settings. Reported results indicate that Sylva can maintain strong robustness under severe heterogeneity (for example, Dirichlet parameter $\alpha=0.1$) while balancing global feature sharing and local adaptation through decoupled LoRA backbone sharing and personalized classifiers.

Two core challenges faced by federated learning in Industrial Internet of Things (IIoT) environments are Non-IID data and poisoning attacks. In IIoT settings, devices generate inherently heterogeneous data, such as distinct operating ranges across different machines, which makes it difficult for traditional federated learning methods to train globally high-performance models. At the same time, malicious participants may

disrupt the learning process by uploading tampered local models or contaminating training data. The Non-IID nature of IIoT data further complicates the distinction between malicious and benign updates.

To address this issue, Xu et al. [70] proposed a deep sign sharing personalized federated learning (DSPFL) scheme. Specifically, during training, the deep neural network models on IIoT devices are partitioned into shallow and deep layers. Gradients corresponding to the shallow layers are updated locally to facilitate personalized feature extraction tailored to the unique data characteristics of each device. For the deep layers, only the signs of the gradients are shared, enabling collaborative learning of complex patterns across devices. The innovation of DSPFL lies in its novel, sole aggregation of the stochastic gradient signs from the deep layers of local models during training. By keeping the shallow layers personalized, DSPFL allows each node to retain unique features critical to its local data distribution, making it difficult for adversaries to craft universally effective poisoned updates for these layers. This targeted aggregation enhances the robustness of the shared components against poisoning attacks, while the shallow layers remain local to preserve personalization. This integrated approach improves the accuracy and resilience of personalized local models on IIoT devices under poisoning attacks.

Experimental results suggest that DSPFL offers clear advantages in highly Non-IID settings. Its personalized model accuracy improves by up to 20% compared with several baselines, while also showing stable convergence. Under model poisoning attacks launched by 30% of nodes, DSPFL maintains an accuracy of 58.38% on the KDD99 dataset, outperforming the second-best method (RobustPFL) by about 3%. Against data poisoning attacks, it achieves a detection accuracy of 47.04% for poisoned categories. Even under adaptive attacks such as STAT-OPT and DNY-OPT, DSPFL sustains an accuracy between 57% and 60%. In addition, DSPFL has favorable computational and communication characteristics because each round shares only the signs of part of the model parameters, which makes it appealing for resource-constrained IIoT environments.

6. Part IV: scenario-specific extensions

Robustness requirements also change substantially across deployment environments. Wireless, edge, and IoT systems introduce communication noise, limited computation, intermittent participation, and stronger physical-world constraints. This part extends the previous taxonomy to these scenario-specific settings and shows how environment-specific assumptions reshape both the threat model and the feasible defense set.

6.1. Wireless and edge federated learning

In wireless and edge deployments, robustness is influenced not only by malicious clients but also by imperfect communication channels, hardware limits, and scheduling uncertainties. This means that classical security taxonomies must be complemented by communication-and-computation co-design concerns.

In practical deployments, federated learning systems must be robust not only to malicious Byzantine behavior but also to environmental uncertainty, which can be viewed as a form of non-adversarial robustness. Beitollahi and Lu [6] noted that when federated learning is deployed at the wireless network edge, physical channel characteristics such as noise, interference, and fading can significantly undermine model convergence. Consequently, research on federated learning robustness in wireless environments mainly focuses on convergence stability under lossy communication and limited resources.

6.2. Robustness against communication noise

Noise in wireless channels leads to inaccurate transmission of model parameters or even transmission failures, posing challenges to the

model's convergence rate and final accuracy. To address this issue, existing research primarily enhances robustness through two approaches: physical layer control and algorithmic adjustment.

- **Dynamic Signal-to-Noise Ratio Scaling:** To maintain the same $O(1/T)$ convergence rate in noisy communication environments as in noise-free environments, the signal-to-noise ratio must be managed dynamically. Research by Wei and Shen [64] demonstrates that by utilizing power control or diversity gain techniques to scale the signal-to-noise ratio of uplink and downlink channels by $O(r^2)$, the impact of noise in the later stages of training can be effectively mitigated. This approach essentially trades the dynamic allocation of physical layer resources for robustness at the algorithmic level.
- **Loss Function Regularization:** Another method to enhance noise robustness involves adjustments at the algorithmic level. Ang et al. [1] proposed adding regularization terms to the loss function to widen its flat regions, thereby reducing the model's sensitivity to noise perturbations. This method allows the model to remain near the optimal solution even when parameters are subject to noise interference.
- **Noise-Aware Scheduling:** When poor channel conditions lead to high packet loss rates that threaten system stability, scheduling strategies based on signal-to-interference-plus-noise ratio can be more robust than random scheduling. Yang et al. [73] proposed prioritizing devices with better channel conditions to ensure the upload of high-quality gradients. Under high-SINR and i.i.d. settings, this policy reduces learning time compared with random and round-robin scheduling; however, its advantage diminishes under low SINR because fewer users contribute their data.

6.3. System robustness under resource constraints and heterogeneity

Wireless edge devices often face issues of limited battery energy and heterogeneous computational capabilities, leading to frequent node dropouts or stragglers, which affect the stability of the global model.

- **Asynchronous Computation and Fault Tolerance:** To address differences in device computational capabilities, asynchronous federated learning allows devices to finish local computation at different time points, eliminating the requirement for synchronized training completion across all devices. For example, Yang et al. [74] jointly optimized time allocation, bandwidth, transmit power, CPU frequency, and learning accuracy under a latency constraint, reducing total energy consumption by up to 59.5% compared with conventional federated learning.
- **Scheduling with Imperfect Channel State Information:** In practical systems, the coordinator is often unable to obtain perfect channel state information. Wadu et al. [60] utilized Gaussian process regression to predict channel information and conducted scheduling based on the Lyapunov optimization framework. In simulations under perfect and imperfect channel state information, the proposed method reduced accuracy loss by up to 25.8% compared with state-of-the-art scheduling and resource-allocation methods.

6.4. Security and privacy enhancement in wireless environments

Although the primary focus is on environmental robustness, the open nature of wireless networks also introduces specific security challenges.

- **Reputation-Based Defense:** Addressing the issue of unreliable or malicious nodes uploading low-quality models, Kang et al. [29] proposed a reputation-based device selection mechanism. By using blockchain to manage reputation scores and rewarding or penalizing devices based on data contribution quality and training behavior, the robustness of the global model is enhanced against poisoning attacks or "free-riding" behaviors.

- **Physical Layer Privacy:** By leveraging the superposition nature of wireless channels and channel noise, a form of natural differential privacy can be achieved. For example, the wireless aggregation scheme studied by Seif et al. [51] combines local differential privacy with bandwidth-efficient aggregation, and its per-user privacy leakage scales as $O(K^{-1/2})$ for K participating users, rather than remaining constant as in orthogonal transmission.

Research by Beitollahi and Lu [6] indicates that the robustness of federated learning in wireless networks is not merely an algorithmic issue but a problem of communication and computation co-design. Future research on robust federated learning should place greater emphasis on physical layer uncertainties, exploring the realization of theoretically guaranteed convergence under imperfect communication conditions. Table 14 specifies a representative deployment or evaluation example for each robustness-enhancement method.

6.5. IoT and IIoT federated learning

As the number of IoT devices proliferates and the amount of data generated is enormous, traditional centralized AI approaches face challenges such as latency, privacy, scalability and energy consumption. Federated learning enables distributed devices to collaboratively train models without sharing raw data, making it particularly suitable for privacy-sensitive IoT scenarios. IoT devices generate massive amounts of data, but centralized processing faces challenges such as privacy leakage, bandwidth constraints, and latency issues. Federated learning addresses these problems through local training and model aggregation [3].

In IoT environments, the most severe security threat to federated learning is poisoning attacks, which compromise the integrity of the training process via malicious clients. These attacks manifest in two primary forms: data poisoning and model poisoning.

- **Data poisoning** involves manipulating local training data. Specifically, label-flipping attacks alter correct labels to incorrect ones, such as marking benign samples as malignant in medical diagnostic data. Meanwhile, backdoor attacks inject hidden triggers into the data. For instance, attackers may use specific pixel patterns in images to induce misbehavior in the model for specific inputs, such as the misidentification of traffic signs in autonomous driving systems.
- **Model poisoning** is more direct: attackers tamper with local model update parameters, such as gradients or model weights, to inject bias into the global model. This may lead to overall performance degradation or even systemic failure, exemplified by the breakdown of anomaly detection in industrial sensor networks.

These attacks are especially harmful in IoT scenarios because resource-constrained distributed devices usually lack strong local validation capabilities, and highly heterogeneous data distributions (Non-IID) make malicious updates easier to hide. Attacks can be categorized into targeted attacks, such as backdoor attacks that affect specific inputs, and untargeted attacks, such as label-flipping attacks that broadly reduce model accuracy. Their success is influenced by client-side vulnerabilities and weaknesses in the aggregation mechanism, especially the sensitivity of FedAvg to outliers. In high-stakes IoT applications, such attacks may amplify model bias or trigger critical failures, such as erroneous load prediction in smart grids [25].

In IoT-based federated learning systems, defenses can be grouped into three directions: secure aggregation, privacy preservation, and malicious-update detection [35]. Secure aggregation schemes, including lightweight secret-sharing variants, protect intermediate updates but may still struggle with Non-IID poisoning. Privacy-preserving methods such as SMPC, homomorphic encryption, differential privacy, and local differential privacy strengthen confidentiality, yet they typically trade accuracy, latency, or energy efficiency for stronger protection. Detection-oriented approaches, including FLTrust-style trust scoring,

Table 14

Representative robustness enhancement methods and examples in wireless federated learning.

Category	Specific Method	Representative Example	Robustness Benefit	Limitation
Communication Robustness	Dynamic SNR Scaling [64]	For strongly convex FL over noisy uplink and downlink channels, scale both SNRs as $\mathcal{O}(t^2)$ as training round t increases.	Preserves the theoretical $\mathcal{O}(1/T)$ convergence rate of noise-free communication.	Requires time-varying power control or diversity gain and increasing physical-layer resources.
	Loss Function Regularization [1]	Under Gaussian communication noise, train a flattened, regularized loss function instead of the original objective.	Improves learning time and reduces parameter-noise sensitivity in noisy networks.	The modified objective may converge to a local optimum.
	Noise-Aware Scheduling [73]	In high-SINR and i.i.d. settings, schedule devices with the best proportional channel conditions in each round.	Achieves shorter learning time than random and round-robin scheduling in the reported setting.	Omits some users' data and can underperform random and round-robin scheduling at low SINR.
System & Resource	Asynchronous Computation [74]	Jointly optimize time, bandwidth, transmit power, CPU frequency, and learning accuracy under a latency constraint.	Reduces total energy consumption by up to 59.5% compared with conventional FL in the reported evaluation.	Requires adjustable computation and communication resources and introduces additional optimization complexity.
	Scheduling with Imperfect CSI [60]	Use Gaussian process regression and Lyapunov optimization for joint client scheduling and resource-block allocation under imperfect CSI.	Reduces accuracy loss by up to 25.8% compared with state-of-the-art methods in the reported simulations.	Channel-prediction errors may lead to suboptimal scheduling decisions.
Security & Privacy	Reputation-Based Defense [29]	In mobile-network FL, use consortium-blockchain reputation based on data contribution and training behavior to select reliable devices.	Reduces the effect of unreliable workers on model accuracy.	Blockchain and reputation management increase system overhead and may affect convergence.
	Physical Layer Privacy [51]	Wireless gradient aggregation exploits the superposition of a Gaussian multiple-access channel to provide local differential privacy.	Combines bandwidth-efficient aggregation with per-user privacy leakage scaling as $\mathcal{O}(K^{-1/2})$.	Requires wireless-resource and privacy-budget control and introduces a convergence-privacy trade-off.

CSI: Channel State Information; i.i.d.: independent and identically distributed; SINR: Signal-to-Interference-plus-Noise Ratio; SNR: Signal-to-Noise Ratio. Examples are representative deployment or evaluation settings from the cited studies; results across rows are not directly comparable.

Table 15

Representative privacy protection techniques.

References	Technique	Advantages	Disadvantages
[35]	SMPC	Strong privacy guarantees	High computation and multi-round communication
[35]	Homomorphic encryption	Supports computation on encrypted data	Limited operation support and large ciphertexts
[22]	Differential privacy	Low computational cost	Injected noise reduces accuracy
[71]	Threshold secure aggregation	Tolerates malicious aggregators and lowers communication cost	Mostly suited to cross-silo settings and still incurs crypto overhead

similarity-based Sybil detection, and anomaly-testing frameworks such as Diffense, can effectively reduce poisoning impact but often require stronger assumptions, additional clean data, or higher server-side computation. Table 15 summarizes representative privacy techniques in this design space.

These techniques rely on multi-dimensional evaluation metrics and validation on heterogeneous datasets, such as Medical MNIST and industrial EDGE-IIoTset. However, they generally face two major bottlenecks: adaptability to Non-IID data and resource consumption [2]. Cyber-physical-social systems make this deployment challenge even sharper because threat detection must be accurate, privacy-preserving, reliable, and explainable to operators. The CPSS model proposed by Yazdinejad et al. combines FL with differential privacy, Paillier-based partially homomorphic encryption, and SHAP explanations, and reports strong anomaly-detection accuracy on SWaT, Gas Pipeline, and UNSW-NB15 datasets [76]. This illustrates why robust FL evaluation in critical infrastructure should include interpretability and operational trust in addition to attack success rate and privacy leakage.

7. Robust federated learning for emerging large-scale AI systems

The transition from task-specific models to foundation models, reinforcement-learning policies, autonomous agents, and

retrieval-augmented systems changes what must be protected. Classical robust FL commonly treats each client contribution as a model-update vector and asks whether the server can aggregate these vectors despite Byzantine or poisoned participants. This abstraction remains useful, but it is no longer sufficient when system behavior also depends on a shared pretrained model, parameter-efficient adapters, environment interactions, persistent agent memory, tool calls, or external retrieval corpora. Table 16 summarizes which existing mechanisms partially transfer and where important coverage gaps remain.

7.1. Foundation models and federated fine-tuning of LLMs

Foundation-model FL typically avoids full-model training because communication and device-memory costs are prohibitive. Parameter-efficient fine-tuning (PEFT) makes federation more feasible by exchanging prompts, adapters, or other small trainable modules instead of all model parameters [32,56,79]. This reduces the communication surface, and existing robust aggregation, secure aggregation, differential privacy, and anomaly filtering can in principle be applied to PEFT updates. However, smaller updates are not automatically easier to secure. A malicious adapter or prompt can induce semantic or instruction-following failures that are difficult to detect through update norms or distances alone, while secure aggregation can conceal the client-level signals needed by anomaly detectors. Moreover, update-level defenses do not validate the provenance or integrity of the shared foundation model itself.

The robustness objective also expands beyond in-distribution accuracy. As discussed in Section 5, federated fine-tuning can distort pretrained representations and reduce out-of-distribution robustness under label and quantity skew [18]. For LLMs, evaluation should additionally cover instruction-following behavior, harmful-output rates, prompt-triggered backdoors, multilingual and domain transfer, and the persistence of attacks after adapter merging. Promising directions include layer- and module-aware robust aggregation for PEFT, semantic validation using diverse server-side probes, signed model and adapter provenance, and privacy mechanisms that preserve enough auditable information to detect malicious fine-tuning.

Table 16

Translation of existing robustness mechanisms to emerging federated AI systems.

Emerging Setting	Expanded Attack Surface	Mechanisms That Partially Transfer	Why Direct Translation Fails	Priority Robustness Directions
Foundation-model FL and LLM fine-tuning [32,56,79]	Shared base-model supply chain; malicious prompts or adapters; semantic backdoors; high-cost validation	Robust and secure aggregation, DP, anomaly filtering, and update clipping	PEFT updates can be small yet behaviorally harmful; vector similarity misses semantic failures; privacy can hide anomaly signals	Module-aware aggregation, semantic probe suites, signed provenance, and evaluation of multilingual and out-of-distribution behavior
Federated reinforcement learning [68,77]	Poisoned observations, rewards, trajectories, or environments; policy-induced distribution shift	Byzantine-robust aggregation, trust scoring, and client sampling	Malicious environments can yield plausible updates; temporal dependence violates common sample assumptions	Trajectory validation, reward auditing, environment-shift detection, and safety-constrained robust FRL
Agentic federated systems [12]	Inter-agent messages, plans, persistent memory, delegation, and tool calls	Authentication, reputation, verifiable protocols, and robust policy aggregation	Harm can occur at runtime without a poisoned training update; trust changes across tasks and tools	Message provenance, auditable memory, capability control, runtime monitoring, and delegation policies
Federated RAG [41,63,82]	Distributed corpora, indexes, retrieval rankings, query routing, and generated citations	Privacy-preserving communication, access control, reputation, and anomaly detection	Model-update defenses do not inspect retrieved evidence; privacy can obstruct evidence auditing	Source provenance, authenticated indexes, robust evidence fusion, retrieval-consistency checks, and end-to-end red teaming

The evidence base is uneven: LLM fine-tuning and FRL have direct federated studies, whereas robust agentic federation and federated RAG remain comparatively early-stage. The listed gaps and priorities synthesize architecture-level implications rather than claiming mature defense coverage.

7.2. Federated reinforcement learning and agentic systems

Federated reinforcement learning (FRL) introduces a feedback loop between the learned policy and the data subsequently collected. Clients may differ not only in data distributions but also in transition dynamics, rewards, exploration policies, and safety constraints. Consequently, malicious behavior can be injected through observations, rewards, or local environments without producing an obviously anomalous model update. Local-environment poisoning can degrade FRL through manipulated observations and rewards, whereas recent provably robust FRL develops ensemble-based robustness guarantees against poisoning attacks [68,77].

Existing robust aggregation remains relevant when attacks manifest as abnormal policy gradients or parameters, but it cannot by itself establish whether trajectories, rewards, or environment interactions are trustworthy. Robust FRL therefore requires trajectory- and reward-level validation, environment-shift detection, safety-constrained policy evaluation, and guarantees that account for temporal dependence rather than assuming independent training samples.

Agentic federated systems extend this challenge further. Federated multi-agent reinforcement learning already enables distributed agents to optimize policies collaboratively without pooling their local training data [12]. More autonomous systems may additionally exchange messages, plans, memories, retrieved evidence, or tool-use traces. In these settings, a client can behave benignly during parameter aggregation yet manipulate shared memory, issue unsafe tool calls, or propagate deceptive messages at inference time. Robustness must therefore combine model-update protection with message provenance, capability and tool-access control, auditable memory, runtime monitoring, and trust policies for inter-agent delegation.

7.3. Federated retrieval-augmented generation

Federated RAG enables multiple parties to collaborate over distributed or private knowledge sources while retaining local control of their corpora [41,63]. This architecture creates a robustness boundary that is largely invisible to conventional FL defenses: the system can be compromised through poisoned documents, malicious retrieval rankings, fabricated evidence, or manipulated query routing even when no client submits a poisoned gradient. PoisonedRAG demonstrates that carefully injected knowledge-base content can steer RAG outputs, illustrating why trustworthy retrieval is an integrity problem rather than merely a model-training problem [82].

Secure aggregation and differential privacy can protect exchanged updates or queries, but they do not verify whether retrieved evidence is accurate or whether a participating corpus is adversarial. Conversely, aggressive filtering may remove rare but valid local knowledge and reduce the value of federation. Robust federated RAG therefore needs document and source provenance, authenticated indexes, robust multi-source evidence fusion, retrieval-consistency checks, privacy-preserving evidence auditing, and end-to-end evaluation that separates retrieval failures from generation failures.

Across these settings, the main methodological shift is from *update robustness* to *workflow robustness*. Future evaluations should jointly test training-time poisoning, inference-time manipulation, privacy leakage, system-resource failures, and compromised external components. They should also report the protected object and trust boundary explicitly: model parameters, adapters, trajectories, agent messages, memory, tools, documents, indexes, or generated evidence. Without this distinction, a method may appear robust while protecting only one component of a larger federated AI pipeline.

8. Future research directions

After organizing the survey around threat models, defense families, cross-cutting trade-offs, scenario-specific extensions, and emerging large-scale AI systems, a consistent pattern emerges: most methods are effective only under a restricted combination of adversarial assumptions, heterogeneity conditions, and deployment constraints. Table 17 therefore summarizes the *typical alignment* between major defense families and major threat categories, rather than claiming universal effectiveness. To make each family concrete, the table also specifies representative methods discussed in this survey. The table is intended as a comparative guide to scope and limitations. In practice, the protection offered by a method depends on whether the evaluation setting matches its trust assumptions, privacy budget, communication model, and attacker strength.

In this section, we synthesize the main unresolved problems highlighted across the four parts and the emerging-system analysis, and outline future directions for more deployable and evidence-grounded robust federated learning.

The discussion above suggests that the next stage of robust federated learning research should move away from isolated optimizations and toward integrated system design. To make this agenda more concrete, Table 18 translates the main unresolved issues into research

Table 17

Typical alignment between defense families and federated learning threats with representative examples.

Defense Family	Representative Examples	Optimal Threat Coverage	Partial Threat Coverage	Main Limitation or Dependency
Robust aggregation and anomaly filtering	AFL, FedMFG, and FedDet [13,58,72]	Byzantine updates, untargeted poisoning, and some backdoor signals	Privacy leakage, coordinated or defense-aware poisoning, severe Non-IID skew, and protocol-level integrity	Often depends on separable benign/malicious updates, trusted reference data, benign-majority assumptions, or limited attacker knowledge
Backdoor-specific defenses	BadCleaner [78]	Targeted poisoning and trigger-based manipulation	General Byzantine failures, coordinated triggers, persistent backdoors, and protocol compromise	Frequently needs clean validation data, server-side distillation, or attack-specific heuristics
Differential privacy and local privacy mechanisms	KD + LDP and wireless local differential privacy [24,51]	Membership inference, gradient inversion, and output-level leakage	Poisoning resilience, fairness, and utility under strong noise	Privacy protection often reduces accuracy or obscures signals used for malicious-update detection
Secure aggregation and cryptographic protection	MASKCRYPT, TAPFed, and encrypted-gradient auditing [27,71,75]	Honest-but-curious servers, update confidentiality, and intermediate leakage	Model replacement, colluding clients, adaptive adversaries, and end-to-end system robustness	Hidden updates can prevent anomaly inspection; confidentiality does not guarantee integrity
Verifiable aggregation and protocol auditing	VERSA [26]	Aggregation correctness, protocol integrity, and model-recovery risks	Data poisoning, client-side backdoors, and subgroup fairness	Requires scalable key management, additional protocol machinery, and careful deployment design
Personalization-aware and heterogeneity-aware defenses	Sylva and DSPFL [46,70]	Non-IID robustness, client drift, and local utility under skew	Global guarantees against strong collusion or universal attacks	Personalized models complicate evaluation and may weaken globally uniform security guarantees

Examples are representative methods discussed in this survey rather than exhaustive members of each defense family; actual coverage depends on the threat model and deployment assumptions.

questions, current gaps, and evaluation priorities. The table emphasizes that a useful roadmap should specify not only *what* to optimize, but also the threat model, evidence standard, resource budget, and deployment boundary under which the claim is expected to hold.

Nine directions appear especially important.

- 1. Threat models beyond independent malicious clients:** Future work should make coalition size, attacker knowledge, coordination channels, update visibility, and adaptation horizon first-class experimental variables. Benchmarks should include coordinated poisoning, distributed triggers, model replacement behind secure aggregation, defense-aware optimization, and cross-round persistence. Defenses should be evaluated against attackers that know their mechanisms and can revise strategies over time, rather than only against fixed attack scripts [4,21,52,66,80].
- 2. Robustness under extreme Non-IID distributions:** Robustness under heterogeneity is not only a matter of lower average accuracy. Under extreme label, quantity, feature, and temporal skew, benign updates may form several distant clusters, and minority-client updates may resemble outliers. This makes the central open problem one of discrimination: a defense must reject malicious updates without suppressing rare but legitimate local distributions. Future work should therefore evaluate defenses under combined skew, partial participation, client drift, and subgroup imbalance, and should report false-positive rates for benign minority clients in addition to global accuracy and attack success [28,42,46,50].
- 3. Joint privacy-robustness-fairness design:** A central open problem is how to combine privacy protection, malicious-update detection, and fairness assurance. Differential privacy, secure aggregation, and encryption reduce leakage risk, but they can also hide the statistical signals that robust aggregation and fairness auditing rely on. Conversely, fairness-aware training may require sensitive-attribute information that is unavailable or should remain private. Future work should study defense stacks that report the full Pareto surface among privacy budget, clean accuracy, attack success rate, fairness gap, latency, and communication cost, rather than optimizing each objective separately [20,22,45,48,53].

- 4. Certified robustness guarantees:** A further challenge is to move beyond empirical resilience and state what is provably protected. Certified robustness for federated learning should specify the attacker budget, poisoned-client fraction, perturbation or trigger model, aggregation rule, client-sampling process, and failure probability. Existing certification studies show that formal guarantees are possible, but they also reveal the difficulty of scaling certificates to adaptive poisoning, severe Non-IID data, partial participation, and privacy-preserving protocols. Future work should pair certificates with stress tests so that a reported guarantee is interpretable under realistic deployment assumptions [11,65,67].
- 5. Trustworthy aggregation under encrypted computation:** Secure aggregation, homomorphic encryption, and MPC can protect update confidentiality, but they do not automatically prove that encrypted updates are well-formed or benign. This creates a tension between inspection and confidentiality: the server may need to clip, score, rank, or reject suspicious updates while being unable to observe them directly. Open problems include privacy-compatible robust selection, verifiable norm or range constraints, encrypted anomaly scoring, dropout-tolerant secure computation, and leakage accounting for auxiliary proofs [9,17,26,27,71,75].
- 6. Benchmark standardization and reporting discipline:** Current empirical evidence is fragmented. Different papers use different datasets, client counts, malicious-client ratios, coalition structures, attacker knowledge assumptions, clean reference data, model sizes, privacy budgets, and system traces, which makes direct comparison difficult. Future benchmarks should include a scenario card covering distribution skew, participation model, attacker capability, defense knowledge, privacy mechanism, fairness metrics, communication cost, computation cost, and edge-resource budget. Frameworks such as LEAF and FedScale show how benchmark infrastructure can make federated experiments more reproducible, but robust FL still needs standardized attack and defense suites that include coordinated backdoors, adaptive poisoning, attack persistence, privacy inference, fairness degradation, and deployment cost [10,28,33,34,39].

Table 18
Forward-looking research roadmap for robust federated learning.

Open Problem	Core Research Question	Why Existing Evidence Is Incomplete	Promising Research Route	Representative Basis
Extreme Non-IID robustness	How can defenses distinguish malicious behavior from legitimate minority-client drift when benign updates are multi-modal, sparse, and unstable?	Many defenses are calibrated on moderate label skew or assume benign and malicious updates remain geometrically separable. Under severe feature, label, quantity, and temporal skew, robust filters can reject rare-but-valid clients or miss attacks hidden inside client drift.	Evaluate joint label/quantity/feature skew, partial participation, and temporal drift; combine personalization-aware aggregation with subgroup-sensitive false-positive analysis and attack-persistence metrics.	[28,42,46,50]
Privacy-robustness-fairness optimization	Can one protocol jointly limit privacy leakage, detect malicious updates, and preserve subgroup fairness without requiring plaintext sensitive attributes or raw updates?	Differential privacy and secure aggregation reduce leakage but can hide anomaly signals; fairness auditing often needs sensitive-attribute statistics; robust filtering may disproportionately exclude under-represented clients.	Formulate multi-objective constraints that report privacy budget, attack success rate, clean accuracy, and fairness gap together; develop privacy-preserving fairness audits and robust debiasing mechanisms.	[20,45,48,53]
Certified robustness guarantees	What attacks can be bounded formally when clients are Non-IID, participation is partial, and attackers may poison data, updates, or backdoor triggers?	Most defenses are empirical and benchmark-specific. Existing certification studies cover important but restricted threat classes, often with assumptions about bounded poisoned clients, norm constraints, or randomized mechanisms.	Report certificates together with empirical stress tests; state the certified attacker budget, aggregation rule, heterogeneity model, and failure probability; extend certificates to adaptive and persistent attacks.	[11,65,67]
Trustworthy encrypted aggregation	How can a server verify clipping, selection, robust aggregation, or anomaly constraints when updates are encrypted or hidden by secure aggregation?	Cryptographic confidentiality does not imply update correctness. Plaintext anomaly detection conflicts with secure aggregation, while homomorphic or MPC-based robust aggregation can add latency, memory cost, and new trust assumptions.	Combine secure aggregation with verifiable constraints, secure robust selection, range or norm proofs, encrypted-gradient auditing, and explicit leakage analysis; evaluate both integrity and cryptographic overhead under dropout.	[9,17,26,27,71,75]
Benchmark standardization	What minimum scenario card should accompany a robustness claim so that results are comparable across papers?	Current studies vary datasets, client counts, skew generation, attacker knowledge, coalition size, privacy budget, system traces, clean reference data, and fairness metrics, making relative claims difficult to interpret.	Build reusable benchmark suites with fixed and extensible scenarios; require reporting of threat model, heterogeneity, privacy, fairness, communication, computation, and failure modes.	[10,28,34,39]
Resource-constrained edge deployment	Which robustness guarantees remain useful when clients have limited memory, battery, bandwidth, intermittent connectivity, and heterogeneous accelerators?	Stronger defenses often add server validation, cryptographic computation, communication rounds, or clean-data requirements. These costs can be infeasible for mobile, wireless, and IoT deployments.	Co-design aggregation, compression, scheduling, and lightweight cryptography; report energy, latency, memory, dropout, and device-level heterogeneity in addition to accuracy and attack success.	[2,3,6,30,40]

The rows are intentionally problem-oriented. They identify unresolved conditions under which current robustness claims often become difficult to compare or deploy.

7. Scenario-aware deployment at the wireless and IoT edge:

Robust federated learning in wireless, edge, and IoT environments must account for lossy channels, stragglers, battery limits, memory pressure, heterogeneous accelerators, and partial participation in addition to malicious behavior. A defense that is accurate on a workstation may be unusable if it requires multiple ciphertext uploads, large validation sets, additional synchronization rounds, or expensive server-side proofs. This calls for communication-and-computation co-design, lightweight cryptographic protocols, adaptive client scheduling, and evaluation on realistic device and network budgets [2,3,6,30,40].

8. Workflow-level robustness for emerging federated AI:

Foundation models, reinforcement-learning policies, agents, and RAG pipelines expand the protected object beyond client model updates. Future work should combine training-time defenses with provenance, semantic validation, trajectory and reward auditing, runtime monitoring, and trustworthy retrieval. Benchmarks should explicitly test attacks against adapters, environments, agent memory and messages, tools, documents, and indexes in addition to conventional model poisoning [32,41,63,68,82].

9. From single defenses to composable defense pipelines:

The literature increasingly shows that no single mechanism is sufficient. A promising direction is to develop composable pipelines that combine architecture-aware aggregation, privacy-preserving communication, verification, explainability, personalization, and

post-training lifecycle mechanisms such as unlearning without breaking scalability. TensorUnlearn illustrates how efficient approximate unlearning and local-regime certificates may become relevant to future federated systems that must satisfy deletion requests or reduce membership-inference risk after deployment [49]. This also requires clearer analysis of when multiple defenses are complementary and when they interfere with one another.

Overall, the future of robust federated learning lies less in proposing isolated defenses and more in understanding compatibility: compatibility between privacy and robustness, between personalization and global guarantees, and between cryptographic protection and resource-constrained deployment. Progress in these directions would make the field more relevant to practical high-stakes applications rather than only to controlled benchmark settings.

9. Conclusion

Robust federated learning remains a rapidly evolving research area because privacy preservation alone is insufficient to guarantee secure and reliable collaborative learning. This survey has reviewed the main robustness challenges in federated learning, including poisoning attacks, privacy leakage, Byzantine behavior, communication uncertainty, and scenario-specific constraints in wireless and IoT environments. It has also shown why threat models must move beyond independent malicious clients to account for collusion, coordinated backdoors, model

replacement behind secure aggregation, defense-aware optimization, and cross-round persistence. We summarized representative defenses spanning robust aggregation, privacy-preserving training, secure aggregation, personalized robustness enhancement, and communication-efficient optimization, while emphasizing their assumptions and limitations. For foundation models, LLM fine-tuning, reinforcement learning, agentic systems, and federated RAG, the protected object expands beyond model updates to include adapters, trajectories, messages, memory, tools, and retrieved evidence. A recurring conclusion across the literature is that no single mechanism simultaneously solves robustness, privacy, fairness, efficiency, certification, and heterogeneity. Future progress will therefore depend on integrated, workflow-level designs that make their assumptions explicit, test extreme Non-IID and adaptive-attack conditions, provide formal guarantees where possible, remain compatible with encrypted computation, and report standardized deployment costs for resource-constrained edge systems. We hope this survey provides a useful reference for understanding the current landscape of robust federated learning and for identifying a clearer research roadmap for future work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] F. Ang, L. Chen, N. Zhao, Y. Chen, W. Wang, F.R. Yu, Robust federated learning with noisy communication, *IEEE Trans. Commun.* 68 (2020) 3452–3464, <https://arxiv.org/abs/1911.00251>, <https://doi.org/10.1109/TCOMM.2020.2979149>.
- [2] O.B. Atia, M.A. Samara, I. Bennis, J. Gaber, A. Abouaissa, P. Lorenz, Securing federated learning in IoT: a survey of attacks, defenses, and frameworks, *IEEE Internet Things J.* (2025), <https://doi.org/10.1109/JIOT.2025.3600617>. Early access.
- [3] E. Baccour, N. Mhaisen, A. Erbad, A.A. Abdellatif, A. Mohamed, M. Hamdi, M. Guizani, Pervasive AI for IoT applications: a survey on resource-efficient distributed artificial intelligence, *IEEE Commun. Surv. Tutor.* 24 (2022) 2366–2418, <https://doi.org/10.1109/COMST.2022.3200740>.
- [4] E. Bagdasaryan, A. Veit, Y. Hua, D. Estrin, V. Shmatikov, How to backdoor federated learning, in: *Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics*, PMLR, 2020, pp. 2938–2948, <https://proceedings.mlr.press/v108/bagdasaryan20a.html>.
- [5] M. Baruch, G. Baruch, Y. Goldberg, A little is enough: circumventing defenses for distributed learning, in: *Advances in Neural Information Processing Systems*, 2019, <https://papers.nips.cc/paper/2019/hash/ec1c59141046cd1866bbcbdfb6ae31d4-Abstract.html>.
- [6] M. Beitollahi, N. Lu, Federated learning over wireless networks: challenges and solutions, *IEEE Internet Things J.* 10 (2023) 14749–14762, <https://doi.org/10.1109/JIOT.2023.3285868>.
- [7] J. Bernstein, J. Zhao, K. Azizzadenesheli, A. Anandkumar, signSGD with majority vote is communication efficient and fault tolerant, in: *International Conference on Learning Representations (ICLR)*, 2019, <https://openreview.net/forum?id=BJxhijAcY7>.
- [8] A.N. Bhagoji, S. Chakraborty, P. Mittal, S. Calo, Analyzing federated learning through an adversarial lens, in: *Proceedings of the 36th International Conference on Machine Learning*, PMLR, 2019, pp. 634–643, <https://proceedings.mlr.press/v97/bhagoji19a.html>.
- [9] K. Bonawitz, V. Ivanov, B. Kreuter, A. Marcedone, H.B. McMahan, S. Patel, D. Ramage, A. Segal, K. Seth, Practical secure aggregation for privacy-preserving machine learning, in: *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2017, pp. 1175–1191, <https://doi.org/10.1145/3133956.3133982>.
- [10] S. Caldas, S.M.K. Duddu, P. Wu, T. Li, J. Konecny, H.B. McMahan, V. Smith, A. Talwalkar, LEAF: a benchmark for federated settings, [arXiv:1812.01097](https://arxiv.org/abs/1812.01097), 2018.
- [11] X. Cao, Z. Zhang, J. Jia, N.Z. Gong, FLCert: provably secure federated learning against poisoning attacks, *IEEE Trans. Inf. Forensics Secur.* 17 (2022) 3691–3705, <https://arxiv.org/abs/2210.00584>, <https://doi.org/10.1109/TIFS.2022.3212174>.
- [12] H.H. Chang, Y. Song, T.T. Doan, L. Liu, Federated multi-agent deep reinforcement learning (Fed-MADRL) for dynamic spectrum access, *IEEE Trans. Wirel. Commun.* 22 (2023) 5337–5348, <https://doi.org/10.1109/TWC.2022.3233436>.
- [13] H. Chen, X. Lv, W. Zheng, AFL: attention-based byzantine-robust federated learning with vector filter, in: *2023 IEEE Global Communications Conference (GLOBECOM)*, IEEE, 2023, pp. 595–600, <https://doi.org/10.1109/GLOBECOM54140.2023.10437587>.
- [14] L. Chen, F. Ang, Y. Chen, W. Wang, Robust federated learning with noisy labeled data through loss function correction, *IEEE Trans. Netw. Sci. Eng.* 10 (2023) 1501–1511, <https://doi.org/10.1109/TNSE.2022.3227287>.
- [15] Cui, J., Wang, R., Si, S., Hsieh, C.J., 2023, Scaling up dataset distillation to ImageNet-1K with constant memory, in: *Proceedings of the 40th International Conference on Machine Learning*, PMLR, pp. 6565–6590. <https://proceedings.mlr.press/v202/cui23e.html>.
- [16] D.R.K. Mary, S. Yarradoddi, N. Victor, T.R. Gadekallu, J. Paek, A game-theoretic federated learning approach for ship detection from aerial images, in: *2023 IEEE Global Communications Conference (GLOBECOM)*, IEEE, 2023, <https://doi.org/10.1109/GLOBECOM54140.2023.10437089>.
- [17] C. Dong, J. Weng, M. Li, J.N. Liu, Z. Liu, Y. Cheng, S. Yu, Privacy-preserving and byzantine-robust federated learning, *IEEE Trans. Dependable Secure Comput.* 21 (2024) 889–904, <https://doi.org/10.1109/TDSC.2023.3264697>.
- [18] M. Du, M. Zhang, Y. Pu, Q. Li, S. Ji, Q. Yin, The risk of federated learning to skew fine-tuning features and underperform robustness, *IEEE Trans. Neural Netw. Learn. Syst.* (2025), <https://doi.org/10.1109/TNNLS.2025.3585063>. Early access.
- [19] B. Becker, R. Kohavi, Adult, Dataset, UCI Machine Learning Repository, 1996, <https://archive.ics.uci.edu/dataset/2/adult>, <https://doi.org/10.24432/C5XW20>.
- [20] Y.H. Ezzeldin, S. Yan, C. He, E. Ferrara, S. Avestimehr, FairFed: enabling group fairness in federated learning, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, 2023, pp. 7494–7502, <https://arxiv.org/abs/2110.00857>, <https://doi.org/10.1609/AAAI.V37I6.25911>.
- [21] M. Fang, X. Cao, J. Jia, N.Z. Gong, Local model poisoning attacks to byzantine-robust federated learning, in: *29th USENIX Security Symposium (USENIX Security 20)*, USENIX Association, 2020, pp. 1605–1622, <https://www.usenix.org/conference/usenixsecurity20/presentation/fang>.
- [22] M.M. Fouda, Z.M. Fadlullah, M.I. Ibrahim, N. Kato, Privacy-preserving data-driven learning models for emerging communication networks: a comprehensive survey, *IEEE Commun. Surv. Tutor.* 27 (2025), <https://doi.org/10.1109/COMST.2024.3486690>.
- [23] X. Fu, S. Lou, J. Zheng, C. Chi, J. Yang, D. Wang, C. Zhu, B. Huang, X. Zhu, Deep learning techniques for DDoS attack detection: concepts, analyses, challenges, and future directions, *Expert Syst. Appl.* 291 (2025) 128469, <https://doi.org/10.1016/j.eswa.2025.128469>.
- [24] G. Gad, Z.M. Fadlullah, M.M. Fouda, M.I. Ibrahim, N. Nasser, Joint knowledge distillation and local differential privacy for communication-efficient federated learning in heterogeneous systems, in: *2023 IEEE Global Communications Conference (GLOBECOM)*, IEEE, 2023, <https://doi.org/10.1109/GLOBECOM54140.2023.10437358>.
- [25] B. Ghimire, D.B. Rawat, Recent advances on federated learning for cybersecurity and cybersecurity for federated learning for internet of things, *IEEE Internet Things J.* 9 (2022) 8229–8249, <https://doi.org/10.1109/JIOT.2022.3150363>.
- [26] C. Hahn, H. Kim, M. Kim, J. Hur, VerSA: verifiable secure aggregation for cross-device federated learning, *IEEE Trans. Dependable Secure Comput.* 20 (2023) 36–52, <https://doi.org/10.1109/TDSC.2021.3126323>.
- [27] C. Hu, B. Li, MASKCRYPT: federated learning with selective homomorphic encryption, *IEEE Trans. Dependable Secure Comput.* 22 (2025) 221–233, <https://doi.org/10.1109/TDSC.2024.3392424>.
- [28] W. Huang, M. Ye, Z. Shi, G. Wan, H. Li, B. Du, Q. Yang, Federated learning for generalization, robustness, fairness: a survey and benchmark, *IEEE Trans. Pattern Anal. Mach. Intell.* 46 (2024) 9387–9406, <https://doi.org/10.1109/TPAMI.2024.3418862>.
- [29] J. Kang, Z. Xiong, D. Niyato, Y. Zou, Y. Zhang, M. Guizani, Reliable federated learning for mobile networks, *IEEE Wirel. Commun.* 27 (2020) 72–80, <https://arxiv.org/abs/1910.06837>.
- [30] S. Kang, Y. Wu, Y. Wang, Q. Wang, X. Liang, Robust federated learning algorithm based on chaotic encryption and dual-server detection, *IEEE Internet Things J.* 12 (2025) 28043–28059, <https://doi.org/10.1109/JIOT.2025.3565598>.
- [31] B. Kar, W. Yahya, Y.D. Lin, A. Ali, Offloading using traditional optimization and machine learning in federated cloud-edge-fog systems: a survey, *IEEE Commun. Surv. Tutor.* 25 (2023) 1199–1226, <https://doi.org/10.1109/COMST.2023.3239579>.
- [32] W. Kuang, B. Qian, Z. Li, D. Chen, D. Gao, X. Pan, Y. Xie, Y. Li, B. Ding, J. Zhou, FederatedScope-LLM: a comprehensive package for fine-tuning large language models in federated learning, in: *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, ACM, 2024, pp. 5260–5271, <https://arxiv.org/abs/2309.00363>, <https://doi.org/10.1145/3637528.3671573>.
- [33] K.N. Kumar, C.K. Mohan, L.R. Cenkramaddi, The impact of adversarial attacks on federated learning: a survey, *IEEE Trans. Pattern Anal. Mach. Intell.* 46 (2024) 2672–2691, <https://doi.org/10.1109/TPAMI.2023.3322785>.
- [34] F. Lai, Y. Dai, S.S. Singapuram, J. Liu, X. Zhu, H.V. Madhyastha, M. Chowdhury, FedScale: benchmarking model and system performance of federated learning at scale, [arXiv:2105.11367](https://arxiv.org/abs/2105.11367), 2021.

- [35] R. Laidi, N. Merabtin, D. Djenouri, S. Latif, H.A. Qadir, Y. Djenouri, I. Balasingham, Federated learning in IoT environments: examining the three-way see-saw for privacy, model-performance, and network-efficiency, *IEEE Commun. Surv. Tutor.* 28 (2026) 1025–1058, <https://doi.org/10.1109/COMST.2025.3557475>.
- [36] J. Larson, S. Mattu, L. Kirchner, J. Angwin, How we analyzed the COMPAS recidivism algorithm, *ProPublica*, 2016, <https://www.propublica.org/article/how-we-analyzed-the-compas-recidivism-algorithm>.
- [37] H.B. Lee, D.B. Lee, S.J. Hwang, Dataset condensation with latent space knowledge factorization and sharing, *arXiv:2208.10494*, 2022.
- [38] S. Lei, D. Tao, A comprehensive survey of dataset distillation, *IEEE Trans. Pattern Anal. Mach. Intell.* 46 (2024) 17–32, <https://doi.org/10.1109/TPAMI.2023.3322540>.
- [39] Y. Li, Z. Guo, N. Yang, H. Chen, D. Yuan, W. Ding, Threats and defenses in the federated learning life cycle: a comprehensive survey and challenges, *IEEE Trans. Neural Netw. Learn. Syst.* 36 (2025) 15643–15663, <https://doi.org/10.1109/TNNLS.2025.3563537>.
- [40] W.Y.B. Lim, N.C. Luong, D.T. Hoang, Y. Jiao, Y.C. Liang, Q. Yang, D. Niyato, C. Miao, Federated learning in mobile edge networks: a comprehensive survey, *IEEE Commun. Surv. Tutor.* 22 (2020) 2031–2063, <https://doi.org/10.1109/COMST.2020.2986024>.
- [41] Q. Mao, Q. Zhang, H. Hao, Z. Han, R. Xu, W. Jiang, Q. Hu, Z. Chen, T. Zhou, B. Li, Y. Song, J. Dong, J. Li, P.S. Yu, Privacy-preserving federated embedding learning for localized retrieval-augmented generation, *arXiv:2504.19101*, 2025.
- [42] Z. Lu, H. Pan, Y. Dai, X. Si, Y. Zhang, Federated learning with Non-IID data: a survey, *IEEE Internet Things J.* 11 (2024), <https://doi.org/10.1109/JIOT.2024.3376548>.
- [43] H.B. McMahan, E. Moore, D. Ramage, S. Hampson, B.A. Y Arcas, Communication-efficient learning of deep networks from decentralized data, in: *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR, 2017, pp. 1273–1282, <https://proceedings.mlr.press/v54/mcmahan17a.html>.
- [44] M.J. Page, J.E. McKenzie, P.M. Bossuyt, I. Boutron, T.C. Hoffmann, C.D. Mulrow, L. Shamseer, J.M. Tetzlaff, E.A. Akl, S.E. Brennan, R. Chou, J. Glanville, J.M. Grimshaw, A. Hrobjartsson, M.M. Lalu, T. Li, E.W. Loder, E. Mayo-Wilson, S. McDonald, L.A. McGuinness, L.A. Stewart, J. Thomas, A.C. Tricco, V.A. Welch, P. Whiting, D. Moher, The PRISMA 2020 statement: an updated guideline for reporting systematic reviews, *BMJ* 372 (2021) n71, <https://doi.org/10.1136/bmj.n71>.
- [45] S. Pentyala, N. Neophytou, A. Nascimento, M.D. Cock, G. Farnadi, PrivFairFL: privacy-preserving group fairness in federated learning, *arXiv:2205.11584*, 2022.
- [46] T. Qi, L. Xue, Y. Zhan, X. Ma, Sylva: tailoring personalized adversarial defense in pre-trained models via collaborative fine-tuning, in: *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security*, ACM, 2025, pp. 1679–1693, <https://doi.org/10.1145/3719027.3744805>.
- [47] C. Qiao, M. Li, Y. Liu, Z. Tian, Transitioning from federated learning to quantum federated learning in internet of things: a comprehensive survey, *IEEE Commun. Surv. Tutor.* 27 (2025) 509–545, <https://doi.org/10.1109/COMST.2024.3399612>.
- [48] B. Rao, J. Zhang, D. Wu, C. Zhu, X. Sun, B. Chen, Privacy inference attack and defense in centralized and federated learning: a comprehensive survey, *IEEE Trans. Artif. Intell.* 6 (2025) 333–353, <https://doi.org/10.1109/TAI.2024.3363670>.
- [49] A.M. Ruzbahani, A. Yazdinejad, H. Karimipour, TensorUnlearn: efficient approximate machine unlearning with Kronecker-factored tensor networks, in: *Poster at the 47th IEEE Symposium on Security and Privacy*, 2026, <https://www.ieee-security.org/TC/SP2026/downloads/posters/sp2026posters-final76.pdf>.
- [50] F. Sattler, S. Wiedemann, K.R. Mueller, W. Samek, Robust and communication-efficient federated learning from non-i.i.d. Data, *IEEE Trans. Neural Netw. Learn. Syst.* 31 (2020) 3400–3413, <https://doi.org/10.1109/TNNLS.2019.2944481>.
- [51] M. Seif, R. Tandon, M. Li, Wireless federated learning with local differential privacy, in: *2020 IEEE International Symposium on Information Theory (ISIT)*, IEEE, 2020, pp. 2604–2609, [arxiv:2002.05151](https://arxiv.org/abs/2002.05151).
- [52] V. Shejwalkar, A. Houmansadr, Manipulating the byzantine: optimizing model poisoning attacks and defenses for federated learning, in: *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, 2021.
- [53] X. Sheng, Z. Yang, W. Bao, FairGuard: a fairness attack and defense framework in federated learning, *IEEE Trans. Dependable Secure Comput.* 22 (2025) 2519–2532, <https://doi.org/10.1109/TDSC.2024.3520134>.
- [54] A. Song, H. Li, K. Cheng, T. Zhang, A. Sun, Y. Shen, Guard-FL: an UMAP-assisted robust aggregation for federated learning, *IEEE Internet Things J.* 11 (2024) 27557–27570, <https://doi.org/10.1109/JIOT.2024.3399259>.
- [55] S.U. Stich, J.B. Cordonnier, M. Jaggi, Sparsified SGD with memory, in: *Advances in Neural Information Processing Systems (NeurIPS)*, 2018, pp. 4447–4458, https://proceedings.neurips.cc/paper_files/paper/2018/hash/b440509a0106086a67bc2ea9df0a1dab-Abstract.html.
- [56] G. Sun, U. Khalid, M. Mendieta, P. Wang, C. Chen, Exploring parameter-efficient fine-tuning to enable foundation models in federated learning, in: *2024 IEEE International Conference on Big Data (BigData)*, IEEE, 2024, pp. 8015–8024, <https://doi.org/10.1109/BIGDATA62323.2024.10825712>.
- [57] H. Sun, T. Zhu, Z. Zhang, D. Jin, P. Xiong, W. Zhou, Adversarial attacks against deep generative models on data: a survey, *IEEE Trans. Knowl. Data Eng.* 35 (2023) 3367–3388, <https://doi.org/10.1109/TKDE.2021.3130903>.
- [58] K. Sun, L. Liu, Q. Pan, J. Li, J. Wu, Large-scale mean-field federated learning for detection and defense: a byzantine robustness approach in IoT, *IEEE Internet Things J.* 11 (2024) 36370–36383, <https://doi.org/10.1109/JIOT.2024.3409610>.
- [59] L.T. Yang, R. Zhao, X. Deng, D. Liu, W. Lu, Tensor-empowered federated learning for cyber-physical-social computing and communication systems, *IEEE Commun. Surv. Tutor.* 25 (2023) 1909–1940, <https://doi.org/10.1109/COMST.2023.3282264>.
- [60] M.M. Wadu, S. Samarakoon, M. Bennis, Federated learning under channel uncertainty: joint client scheduling and resource allocation, in: *Proceedings of the IEEE Wireless Communications and Networking Conference (WCNC)*, IEEE, 2020, pp. 1–6, [arxiv:2002.00802](https://arxiv.org/abs/2002.00802).
- [61] O.A. Wahab, A. Mourad, H. Otok, T. Taleb, Federated machine learning: survey, multi-level classification, desirable criteria and future directions in communication and networking systems, *IEEE Commun. Surv. Tutor.* 23 (2021) 1342–1397, <https://doi.org/10.1109/COMST.2021.3058573>.
- [62] Y. Wan, Y. Qu, W. Ni, L. Gao, Y. Xiang, E. Hossain, Data and model poisoning backdoor attacks on wireless federated learning, and the defense mechanisms: a comprehensive survey, *IEEE Commun. Surv. Tutor.* 26 (2024) 1861–1897, <https://doi.org/10.1109/COMST.2024.3361451>.
- [63] P. Addison, M.T.H. Nguyen, T. Medan, J. Shah, M.T. Manzari, B. McElrone, L. Lalwani, A. More, S. Sharma, H.R. Roth, I. Yang, C. Chen, D. Xu, Y. Cheng, A. Feng, Z. Xu, C-FedRAG: a confidential federated retrieval-augmented generation system, *arXiv:2412.13163*, 2024.
- [64] X. Wei, C. Shen, Federated learning over noisy channels: convergence analysis and design examples, *IEEE Trans. Cogn. Commun. Netw.* 8 (2022) 1253–1268, <https://doi.org/10.1109/TCCN.2022.3140788>.
- [65] C. Xie, M. Chen, P.Y. Chen, B. Li, CRFL: certifiably robust federated learning against backdoor attacks, *arXiv:2106.08283*, 2021.
- [66] C. Xie, K. Huang, P.Y. Chen, B. Li, DBA: distributed backdoor attacks against federated learning, in: *International Conference on Learning Representations*, 2020, <https://openreview.net/forum?id=rkgy50Vfvr>.
- [67] C. Xie, Y. Long, P.Y. Chen, Q. Li, S. Koyejo, B. Li, Unraveling the connections between privacy and certified robustness in federated learning against poisoning attacks, in: *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security*, ACM, 2023, pp. 1511–1525, <https://arxiv.org/abs/2209.04030>, <https://doi.org/10.1145/3576915.3623193>.
- [68] M. Fang, X. Wang, N.Z. Gong, Provably robust federated reinforcement learning, in: *Proceedings of the ACM Web Conference 2025*, ACM, 2025, pp. 896–909, <https://arxiv.org/abs/2502.08123>, <https://doi.org/10.1145/3696410.3714728>.
- [69] Z. Xing, Z. Zhang, Z. Zhang, Z. Li, M. Li, J. Liu, Z. Zhang, Y. Zhao, Q. Sun, L. Zhu, G. Russell, Zero-knowledge proof-based verifiable decentralized machine learning in communication network: a comprehensive survey, *IEEE Commun. Surv. Tutor.* (2025), <https://doi.org/10.1109/COMST.2025.3561657>. Early access.
- [70] C. Xu, N. Sohrabi, Y. Qu, H. Dong, Z. Tari, X. Yi, DSPFL: a deep-layer sign sharing personalized federated learning scheme for mitigating poisoning attacks, *IEEE Trans. Neural Netw. Learn. Syst.* (2025), <https://doi.org/10.1109/TNNLS.2025.3601078>. Early access.
- [71] R. Xu, B. Li, C. Li, J.B.D. Joshi, S. Ma, J. Li, TAPFed: threshold secure aggregation for privacy-preserving federated learning, *IEEE Trans. Dependable Secure Comput.* 21 (2024) 4309–4323, <https://doi.org/10.1109/TDSC.2024.3350206>.
- [72] H. Yang, D. Gu, J. He, A robust and efficient federated learning algorithm against adaptive model poisoning attacks, *IEEE Internet Things J.* 11 (2024) 16289–16302, <https://doi.org/10.1109/JIOT.2024.3351371>.
- [73] H.H. Yang, Z. Liu, T.Q.S. Quek, H.V. Poor, Scheduling policies for federated learning in wireless networks, *IEEE Trans. Commun.* 68 (2020) 317–333, <https://arxiv.org/abs/1908.06287>.
- [74] Z. Yang, M. Chen, W. Saad, C.S. Hong, M. Shikh-Bahaei, Energy efficient federated learning over wireless communication networks, *IEEE Trans. Wirel. Commun.* 20 (2021) 1935–1949, <https://arxiv.org/abs/1911.02417>.
- [75] A. Yazdinejad, A. Dehghantanha, H. Karimipour, G. Srivastava, R.M. Parizi, A robust privacy-preserving federated learning model against model poisoning attacks, *IEEE Trans. Inf. Forensics Secur.* 19 (2024) 6693–6708, <https://doi.org/10.1109/TIFS.2024.3420126>.
- [76] A. Yazdinejad, Z.D. Mohammadabadi, A. Dehghantanha, G. Srivastava, An explainable and privacy-preserving federated learning model for threat detection in cyber-physical-social systems, *IEEE Trans. Comput. Soc. Syst.* (2025), <https://doi.org/10.1109/TCSS.2025.3606798>. Early access.
- [77] E. Ma, P. Rathi, S.R. Etesami, Local environment poisoning attacks on federated reinforcement learning, *arXiv:2303.02725*, 2023.
- [78] J. Zhang, C. Zhu, C. Ge, C. Ma, Y. Zhao, X. Sun, B. Chen, BadCleaner: defending backdoor attacks in federated learning via attention-based multi-teacher distillation, *IEEE Trans. Dependable Secure Comput.* 21 (2024) 4559–4573, <https://doi.org/10.1109/TDSC.2024.3354049>.
- [79] J. Zhang, S. Vahidian, M. Kuo, C. Li, R. Zhang, T. Yu, Y. Zhou, G. Wang, Y. Chen, Towards building the federated GPT: federated instruction tuning, *arXiv:2305.05644*, 2023.
- [80] Z. Zhang, A. Panda, L. Song, Y. Yang, M. Mahoney, P. Mittal, R. Kannan, J. Gonzalez, Neurotoxin: durable backdoors in federated learning, in: *Proceedings of the 39th International Conference on Machine Learning*, PMLR, 2022, pp. 26429–26446, <https://proceedings.mlr.press/v162/zhang22w.html>.
- [81] L. Zhu, Z. Liu, S. Han, Deep leakage from gradients, in: *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 14747–14756, <https://proceedings.neurips.cc/paper/2019/hash/60a6c4002cc7b29142def8871531281a-Abstract.html>.
- [82] W. Zou, R. Geng, B. Wang, J. Jia, PoisonedRAG: knowledge poisoning attacks to retrieval-augmented generation of large language models, in: *34th USENIX Security Symposium (USENIX Security 25)*, 2025, <https://www.usenix.org/conference/usenixsecurity25/presentation/zou>.